Canine Rehabilitation and Physical Therapy

SECOND EDITION

Canine Rehabilitation and Physical Therapy

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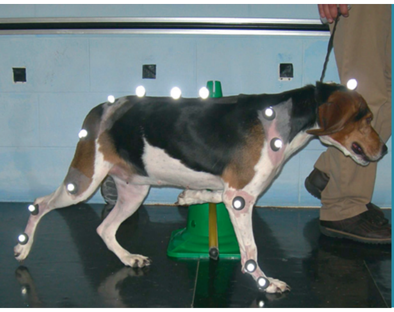

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Laser Therapy in Canine Rehabilitation
Other Modalities in Veterinary Rehabilitation
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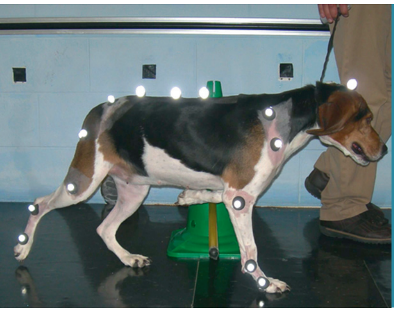


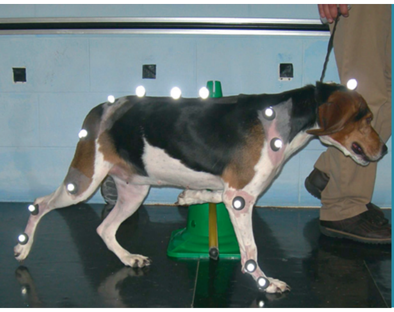
Preface

Individuals involved in veterinary medicine and surgery have witnessed remarkable progress in their ability to diagnose and treat problems that, in the recent past, would have been untreatable. Total hip replacement in the dog, laparoscopic procedures, arthroscopic surgery and the widespread use of sophisticated imaging modalities such as MRI, CT, and real-time ultrasonic imaging are daily occurrences in veterinary medicine. With the availability of these sophisticated diagnostic techniques, the expectations for enhanced functional results have grown.

Human physical therapy is an internationally recognized discipline, and the positive efforts of post-surgical and post-injury rehabilitation have been documented and recognized in human health care. Relatively little attention has been given to veterinary patients afflicted with similar conditions. There is profound interest on the part of veterinary caregivers to learn about and provide rehabilitation and therapy following surgery, illness, or injury. Techniques used in human physical therapy are being adapted for use in small animal patients.

The purpose of this textbook is to provide a solid understanding of physical therapy techniques and interventions for dogs. Most professional programs in veterinary medicine do not provide training in physical therapy. Educational programs in physical therapy do not include evaluation and treatment of animals. These factors have resulted in the need for close collaboration between veterinarians, physical therapists, and veterinary technicians to provide optimal evaluation and treatment of animal patients. Currently, these collaborative efforts and relationships are being used to create treatment protocols for many types of cutaneous, neurologic and musculoskeletal injuries in animals. In this book, we hope to amalgamate the knowledge that physical therapists, veterinarians, and veterinary technicians possess to facilitate a faster and more complete recovery from debilitating conditions.

It is with these goals in mind that this edition was conceived and written. We have seen a tremendous interest, and growth, in veterinary rehabilitation since the first edition of this book was published in 2004. Canine rehabilitation has grown from a small area of interest to a recognized discipline during this time, and all animal patients, their owners, and veterinary caregivers are realizing the benefits of this progression. This book has expanded greatly from the first edition as knowledge within this field has rapidly grown. Many chapters have been added to disseminate this new knowledge, and all of the chapters have been updated to reflect advances in the respective topics. We hope this book is useful as both an in-depth study of the field, and as a quick reference on topics such as protocols for specific conditions that are seen clinically.



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I am truly amazed at the continued interest and progress made in veterinary rehabilitation during the ten years since the first edition of this textbook was published. When writing a textbook in an area that incorporates two separate professions, there are many challenges to overcome and people to thank. I would like to especially thank Dave for his humor, expertise, and continual encouragement to keep things on track. Thanks are also in order to the many physical therapists, veterinarians, veterinary technicians, owners, and patients who have helped move the field forward and provided kind words and encouraged our efforts. The staff at Elsevier, especially Penny and Shelly, have done a tremendous job in keeping things on track and giving us nudges, pushes and shoves when needed. I would be remiss if I did not thank my colleagues at the University of Tennessee for providing help and support in the clinics and laboratory, especially Drs. Joe Weigel, Marti Drum, Dave Hicks, and Jason Headrick. I would especially like to thank my family for continuing to support and encourage me; my wife Linda who has always encouraged and supported my efforts, and put up with the hectic work and travel schedule; Chris and Nick, who are not only my sons, but my best friends, who always keep life interesting; and my mother Adeline and late father, George, who made my dream of becoming a veterinarian come true by encouraging and supporting me from the very beginning. Finally, I would like to thank my patients, past and present, for their wisdom in teaching me about rehabilitation.

All things bright and beautiful,
All creatures great and small,
All things wise and wonderful:
The Lord God made them all.
Cecil Frances Alexander, 1848

Darryl Millis

Many thanks are owed when a book five plus years in the making is finally completed. So many people have generously given their time and thoughts to help me, and all cannot be listed; however, a few special people need to be thanked by name for their support and encouragement. Thanks first to my co-editor and friend, Darryl, who is not only an exceptional surgeon but also an exceptional person. To the many contributors who have helped to keep this work spirited and always enjoyable. My colleagues, students, and patients, I continue to learn every day from all of you. Old friends, like Kevin, Ivan, Denis, and Randy who have kept me laughing for the last 25 plus years, let's keep the good times coming! To many family members across this country who remind me of what family means, and to the memory of my father, Jacob Levine (1913-2011). To my mother Marie, to whom I owe much more than I can ever repay. To my children Lauren Allyn, Sarah Marie, Hadley Christian, and Ava Katherine Ann who brighten every moment of every day. To my wife Allison—who somehow understands me completely and supports me wholly. And to the One who was with me before I was born.

David Levine

SECTION I

Introduction to Physical Rehabilitation

1 History of Canine Physical Rehabilitation

Lin McGonagle, Linda Blythe, and David Levine

History of Canine Physical Rehabilitation

The idea of applying rehabilitation principles and techniques to animals, although not new, has grown appreciably since the mid 1990s. More than 110 facilities providing physical therapy and rehabilitation are now operating in the United States and this number is growing rapidly as veterinarians and physical therapists realize the need and market for these services. Although many of the treatment protocols for humans were developed and continue to be developed using animal models,[1-7] a growing number of research studies are being conducted in universities and private practices that look specifically at the benefits of different methods of rehabilitation in animals, especially dogs. Higher owner expectations combined with increased sophistication and technical abilities of veterinary clinicians have resulted in greater interest in physical therapy and rehabilitation.

Interest in the practice of canine rehabilitation in the United States first gained momentum in the late 1980s and throughout the 1990s as a result of the influence of classic texts; sporadic journal articles; national presentations at the American Physical Therapy Association (APTA), American Veterinary Medical Association (AVMA), and the American College of Veterinary Surgeons (ACVS) meetings; and the formation of the Animal Physical Therapist Special Interest Group within the APTA. The International Racing Greyhound Symposium in conjunction with the Eastern States Veterinary Conference (currently the North American Veterinary Conference) in Orlando, Florida, was first started in 1986 and was expanded and renamed the International Canine Sports Medicine Symposium to include all sporting dogs. Rehabilitation was a frequent topic at these annual meetings and numerous articles have been printed in the proceedings of this symposium as well as others during the past 25 years.[8-19] Many veterinarians and physical therapists have lectured and presented continuing education and research findings at regional, national, and international human and veterinary conferences.

The authors would like to acknowledge Robert Taylor for his work on the previous edition.

Many veterinarians have felt a need to improve postoperative patient care, because traditionally preoperative management, diagnostic procedures, and surgical treatment have been emphasized. The results seen with humans undergoing intensive postoperative rehabilitation have caused many veterinarians to rethink patient management strategies, so that postoperative rehabilitation, once overlooked, is now becoming more common in veterinary practice.

The APTA position statement and the AVMA "Guidelines for Alternative and Complementary Veterinary Medicine" have provided some initial guidelines for the field of animal physical therapy. Each professional organization has recognized the other and has published guidelines for collaborative working relationships. The APTA House of Delegates adopted a position statement in June 1993 regarding animal physical therapy, which states that the APTA "endorses the position that physical therapists may establish collaborative, collegial relationships with veterinarians for the purposes of providing physical therapy services or consultation."[20] The "Guidelines for Alternative and Complementary Veterinary Medicine" were adopted in July 1996 by the AVMA House of Delegates.[21] The document, which defined *veterinary physical therapy* as "the use of noninvasive techniques, excluding veterinary chiropractic, for the rehabilitation of injuries in non-human animals," established the following guidelines:

> Veterinary physical therapy should be performed by a licensed veterinarian or, where in accordance with state practice acts, by (1) a licensed, certified, or registered veterinary or animal health technician educated in veterinary physical therapy or (2) a licensed physical therapist educated in non-human animal anatomy and physiology.

> Veterinary physical therapy performed by a non-veterinarian should be performed under the supervision of, or referral by, a licensed veterinarian who is providing concurrent care. Veterinary physical therapy performed by non-veterinarians should be limited to the use of stretching; massage; stimulation by use of low-level lasers, electrical sources, magnetic fields, and ultrasound; rehabilitative exercises; hydrotherapy; and applications of heat and cold.

New guidelines were adopted by the AVMA House of Delegates in 2001.[22] They evaluated several medical approaches described by the terms *complementary, alternative,* and *integrative* and collectively described them as *complementary and alternative veterinary medicine* (CAVM). Examples of CAVM include aromatherapy; Bach flower remedy therapy; energy therapy; low-energy photon therapy; magnetic field therapy; orthomolecular therapy; veterinary acupuncture, acutherapy, and acupressure; veterinary homeopathy; veterinary manual or manipulative therapy (similar to osteopathy, chiropractic, or physical medicine and therapy); veterinary nutraceutical therapy; and veterinary phytotherapy.

The basic concept behind these new guidelines was to emphasize that CAVM should be held to the same standards as traditional veterinary medicine, including validation of safety and efficacy by the scientific method. In addition, the guidelines state, "The AVMA believes veterinarians should ensure that they have the requisite skills and knowledge for any treatment modality they may consider using." Finally, another pertinent point is, "The quality of studies and reports pertaining to CAVM varies; therefore, it is incumbent on a veterinarian to critically evaluate the literature and other sources of information. Veterinarians and organizations providing or promoting CAVM are encouraged to join with the AVMA in advocating sound research necessary to establish proof of safety and efficacy." It is encouraging that a number of studies have already indicated its benefit in treating a wide number of conditions.[23-34] In addition, other studies have evaluated therapeutic modalities and the responses of tissues to rehabilitation following injury and during repair.[35-97]

Journals and books have provided information on animal physical rehabilitation for more than 30 years. However, an article titled "Postsurgical Physical Therapy: The Missing Link," by Taylor,[23] was one of the first to capture the interest of the veterinary community. Throughout the remainder of the 1990s and continuing to date, the number of publications increased. Topics included cranial cruciate ligament rupture and rehabilitation,[23-34] postoperative management of spinal surgery or spinal cord diseases in the dog,[35-48] orthopedic conditions and osteoarthritis,[49-61] and management considerations for trauma patients.[62,63] Collaboration of veterinarians and physical therapists has increased, resulting in publications regarding physical rehabilitation for the critically ill patient[64-66] and pain relief.[67-70] It is clear that the dissemination of information has strengthened the ties between veterinarians and physical therapists and increased support for the efficacy of rehabilitation approaches in animal care.

Initially, several books helped shape the field of animal physical rehabilitation. *Physical Therapy for Animals: Selected Techniques,* by Downer,[71] influenced professionals as early as 1978. *Canine Sports Medicine and Surgery*[72] added to the growing field of sports medicine and discussed the role of physical rehabilitation in treating injuries of dogs. Originally published in 1994 as *Care of the Racing Greyhound*[74] and revised in 2007,[73] *Care of the Racing and Retired Greyhound* outlines rehabilitation guidelines for musculoskeletal injuries sustained in racing and training that are applicable to any athletic dog.[74] The revised version was expanded to include retired greyhounds. Three recent texts written through the cooperation of veterinarians and physical therapists include *Canine Rehabilitation and Physical Therapy* (2004) by Millis, Levine, and Taylor[75]; *Essential Facts of Physiotherapy* (2004) by Bockstahler, Levine, and Millis[76]; and *Animal Physiotherapy—Assessment, Treatment and Rehabilitation of Animals* (2007) by McGowan, Goff, and Stubbs.[77] All of these recent texts provide details of examination procedures, intervention, and applications for specific conditions. Multiple articles describe the use of physical therapeutic modalities and orthotics.[78-97]

Physical rehabilitation is gaining greater acceptance in veterinary medicine and there are more options for training. Currently, rehabilitation rotations, electives, and instruction are available at a number of veterinary colleges, and rehabilitation lectures and even courses exist as a part of the professional curriculum. A university-based certificate program at the University of Tennessee has over 850 graduates worldwide as of 2012 (www.canineequinerehab.com). Graduates receive the Certified Canine Rehabilitation Practitioner designation. Other continuing education programs are available at conferences and meetings.

In August 1999 the First International Symposium for Physical Therapy and Rehabilitation in Veterinary Medicine was sponsored by and held at Oregon State University. More than 300 participants from 21 countries met during 4 days to present clinical and research findings and to share ideas. This meeting focused entirely on animal physical rehabilitation and brought the professions of veterinary medicine and physical therapy together to exchange information and share ideas. Subsequently, there have been additional symposia in the United States in Knoxville, Tennessee, in 2002; in Raleigh, North Carolina, in 2004; and in Minneapolis, Minnesota, in 2008. The fourth symposium was held in Arnhem, the Netherlands, in 2006 to expand and integrate rehabilitation knowledge, methods, and practices of other countries. In 2010, the sixth symposium was held at Auburn University in Alabama and the Seventh Symposium was in Vienna, Austria. Plans for a meeting in the United States in the summer of 2014 are ongoing.

Worldwide Animal Rehabilitation and Physical Therapy Associations

Veterinarians and physical therapists in many countries have been sharing information and working together for three decades. Physiotherapists have professional organizations for animal physical rehabilitation in at least 11 countries: Australia, Canada, Finland, Germany, the Netherlands, New Zealand, South Africa, Sweden, the United Kingdom, and the United States. In many of these countries, the groups are formally recognized by their respective national physical therapy associations. Box 1-1 provides a brief history of these organizations and contact information.

With the robust interest in the area of canine physical rehabilitation at local, national, and international meetings, there are now several formal veterinary associations. The International Association of Veterinary Rehabilitation and Physical Therapy (www.iavrpt.org) became an official association in July 2008. This diverse group of veterinarians, physical therapists, and other interested professionals grew out of the original symposium attendees and assists in guiding the organization of the biannual symposium. Their mission is to provide a forum for the presentation of

Box 1-1 Animal Physical Rehabilitation Organizations (Official Names)

International Associations

International Association of Veterinary Rehabilitation and Physical Therapy

Incorporated in 2008 for numerous professions working in animal rehabilitation

Website: www.iavrpt.org

Veterinary European Physical Therapy and Rehabilitation Association

First meeting held in 2010 for professions working in animal rehabilitation in Europe.

Website: www.vepra.eu

International Association of Physical Therapists in Animal Practice

Recognized in 2011 by the World Confederation for Physical Therapy

Website: www.WCPT.org

Australia

Animal Physiotherapy Group

Recognized by the Australian Physiotherapy Association.

Website: www.physiotherapy.asn.au

Canada

Animal Rehab Division, formerly known as Canadian Horse and Animal Physiotherapy Association (CHAP)

First organized in 1994. Recognized in 2004 by the Canadian Physiotherapy Association (www.physiotherapy.ca/)

Website: www.physiotherapy.ca/Division/Animal-Rehabilitation

Finland

Finnish Association of Animal Physiotherapists

Founded in 1997. Recognized as a subgroup of the Finnish Association of Physiotherapists

Website: www.fysioterapia.net

The Netherlands

Nederlandsee Verening voor Fysiotherapie bij Dieren (NVFD; Dutch Animal Physical Therapy Association)

Founded in 1989. Recognized by the Dutch Ministry of Agriculture since 1992.

Address: Hindelaan 56, 1216 CW Hilversum, Netherlands

South Africa

Animal Physiotherapy Group of South Africa, formerly known as the South African Association of Physiotherapists in Animal Therapy (SAAPAT)

First organized in 1988. Gained official recognition in 1998 by the South African Society of Physiotherapy.

Website: www.animalphysiogroup.co.za

Sweden

The Association of Registered Physiotherapists of Veterinary Medicine

Founded in 1995. Became a section member of the Swedish Association of Registered Physiotherapists in 1996.

Website: lsvet.se

United Kingdom

Association of Chartered Physiotherapists in Animal Therapy (ACPAT)

Recognized in 1988 by the Chartered Society of Physiotherapists

Website: www.acpat.org

Ireland

Chartered Physiotherapists in Veterinary Practice

Recognized by the Irish Society of Chartered Physiotherapists

Website: iscp.ie

Switzerland

Schweizerischer verband fur tierphysiotherapie

Established in 2007

Website: svtpt.ch, tierphysiotherapie.com

United States

Animal Physical Therapist Special Interest Group

Recognized in 1998 by the American Physical Therapy Association

Website: www.orthopt.org/sig_apt.php

American Association of Rehabilitation Veterinarians

Founded in 2008 for veterinarians.

Website: www.rehabvets.org

American Association of Rehabilitation Veterinary Technicians

Created for veterinary technicians who provide assistance in physical rehabilitation.

http://rehabvets.org/aarvt.lasso

American College of Veterinary Sports Medicine and Rehabilitation.

Provisional approval in 2010 by the American Veterinary Medical Association

Website: VSMR.org

clinical and research information and discussion of topics related to animal rehabilitation, to further scientific investigation, and to promote the continued development of this specialty area to provide improved quality of care based on sound evidence. Another new association, the Veterinary European Physical Therapy and Rehabilitation Association, held its first meeting in 2010 in Zagreb, Croatia. Another organization for veterinarians interested in rehabilitation in the United States is the American Association of Rehabilitation Veterinarians (www.rehabvets.org). This group has a goal of making veterinarians in the United States more aware of the benefits of rehabilitation and addressing practice concerns. A similar organization exists for veterinary technicians, the American Association of Rehabilitation Veterinary Technicians. Another formal association became established in April 2011 within the World Confederation of Physical Therapy (WCPT.org). The International Association of Physical Therapists in Animal Practice was created to foster collaboration among physical therapists worldwide and to share information relating to research, education, and practice.

Future Trends

Physical rehabilitation in small animal practice has become increasingly common and will continue to become more accepted as the scientific literature continues to evolve. In many orthopedic and neurologic conditions, physical rehabilitation is becoming commonplace as a means to enhance recovery, as it is in human medicine and surgery. Wellness and preventive medicine, such as physical rehabilitation for weight reduction and for maintenance of muscle strength and cardiorespiratory fitness, is also emerging as a trend among pet owners. In 2008, a national committee of veterinarians in the United States proposed that veterinary rehabilitation become a board-certified specialty under the American Veterinary Medical Association and the American Board of Veterinary Specialties (ABVS). This newly formed American College of Veterinary Sports Medicine and Rehabilitation has been given provisional approval by the AVMA to establish and maintain credentialing and certification standards for veterinary practitioners who excel in sports medicine and rehabilitation. The organization formed with a group of charter diplomats and administered its first certifying examination in 2012. With both academic and nonacademic pathways to certification, a veterinarian can become board certified in either canine or equine specialties under this college.

It is certain the field of animal rehabilitation will expand in future years both in the United States and worldwide. Our hope is that veterinary and physical therapy professionals will continue to collaborate to define practice models and establish effective clinical protocols, as well as provide education and training for the benefit of our animal clients and owners.

REFERENCES

1. Daily L et al: The effects of microwave diathermy on the eye of the rabbit, *Am J Ophthalmol* 35:1001-1017, 1952.
2. Douglas WW, Malcolm JL: The effect of localized cooling on conduction of cat nerves, *J Physiol* 130:53-71, 1955.
3. Lehman JF et al: Ultrasonic effects as demonstrated in live pigs with surgical metal implants, *Arch Phys Med Rehabil* 40:483-488, 1959.
4. Nadasdi M: Inhibition of experimental arthritis by athermic pulsing shortwave in rats, *Am J Orthop* 2:105-107, 1960.
5. Ely TS et al: Heating characteristics of laboratory animals exposed to ten centimeter microwaves, *Biol Eng* 11:123-137, 1964.
6. Jezdinsky J et al: Effects of cold and heat therapy on traumatic oedema of rat hind paw, *Acta Univ Palacki Olomuc Fac Med* 66:185-201, 1973.
7. Michalski WJ, Sequin J: The effect of muscle cooling and stretch on muscle spindle secondary endings in the cat, *J Physiol* 253:341-356, 1975.
8. Van Dyke J: Canine rehabilitation—therapeutic protocols for specific sports injuries, *Proceedings of the Twenty-fifth International Canine Sports Medicine Symposium,* Jan 17-18, Orlando, Fla, 2009.
9. Dee JF: It's just a sprain, *Proceedings of the Fourteenth International Canine Sports Medicine Symposium*, Jan 10-11, Orlando, Fla, 28, 1998.
10. Taylor RA: First aid and bandage/splint techniques for sporting dogs, *Proceedings of the Thirteenth International Canine Sports Medicine Symposium,* Jan 14-15, Orlando, Fla, 38, 1997.
11. Taylor RA: Rehabilitation of injuries, *Proceedings of the Thirteenth International Canine Sports Medicine Symposium,* Jan 11-12, Orlando, Fla, 45, 1997.
12. Taylor RA: Basic principles of physical therapy for sporting dogs, *Proceedings of the Eleventh International Canine Sports Medicine Symposium,* Jan 14-15, Orlando, Fla, 43, 1995.
13. Dumbleton CD: Electro therapy and the racing greyhound, *Proceedings of the Tenth International Racing Greyhound Symposium,* Jan 15-16, Orlando, Fla, 1994.
14. deHaan JJ et al: Double blind clinical study of PSGAG's (Adequan) for treatment of hip dysplasia in the adult dogs, *Proceedings of the Ninth Annual International Racing Greyhound Symposium,* Jan 16-17, Orlando, Fla, 18, 1993.
15. Sweeney PA: Application of physiotherapy in the racing greyhound, *Proceedings of the Ninth International Racing Greyhound Symposium,* Jan 16-17, Orlando, Fla, 37, 1993.
16. Taylor RA: Practical bandaging and splinting, *Proceedings of the Ninth International Racing Greyhound Symposium,* Jan 16-17, Orlando, Fla, 39, 1993.
17. Taylor RA: Management of digit injuries, *Proceedings of the Eighth International Racing Greyhound Symposium Proceedings,* Jan 11-12, Orlando, Fla, 17, 1992.
18. Starbuck MA: Physical therapy for racing greyhounds, *Proceedings of the Seventh International Racing Greyhound Symposium,* Jan 12-13, Orlando, Fla, 46, 1991.
19. Taylor RA: Bandaging techniques for common racing injuries, *Proceedings of the Seventh International Racing Greyhound Symposium,* Jan 12-13, Orlando, Fla, 51, 1991.

20. Position on physical therapists in collaborative relationships with veterinarians, *American Physical Therapy Association House of Delegates* 06-93-20-36(Program 32), 1993.
21. Guidelines for alternative and complementary veterinary medicine. In *AVMA directory*, Schaumburg, Ill, 2000, American Veterinary Medical Association.
22. Guidelines for complementary and alternative veterinary medicine. In *AVMA policy statements and guidelines*, Schaumberg, Ill, 2001, American Veterinary Medical Association.
23. Taylor RA: Postsurgical physical therapy: the missing link, *Compend Cont Educ Pract Vet* 14:1583-1594, 1992.
24. Johnson JM et al: Rehabilitation of dogs with surgically treated cranial cruciate ligament-deficient stifles by use of electrical stimulation of muscles, *Am J Vet Res* 58:1473-1477, 1997.
25. Millis DL et al: A preliminary study of early physical therapy following surgery for cranial cruciate ligament rupture in dogs, *Vet Surg* 26:434, 1997.
26. Marcellin-Little DJ: Cranial cruciate ligament injuries—treatment decisions, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 133, 2004.
27. Millis DL: Cranial cruciate ligament injuries—rehabilitation, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 137, 2004.
28. Adamson CP: Cranial cruciate ligament surgery, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 215, 2002.
29. Adamson CP, Taylor RA: General outcomes twelve weeks post tibial plateau leveling osteotomy in dogs with cranial cruciate ligament injury, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 205, 2002.
30. Francis DA et al: Bone and muscle loss from disuse following cranial cruciate ligament transection and stifle stabilization surgery, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 203, 2002.
31. Bockstahler B et al: The effect of physical therapy on postoperative rehabilitation of dogs after cranial cruciate ligament repair, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 201, 2002.
32. Taylor RA et al: Stifle surgery and rehabilitation, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 143, 2002.
33. Evans CS: A rehabilitation protocol following tibial plateau leveling osteotomy of the canine stifle: a single subject case report, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 135, 2002.
34. Millis DL: Changes in muscle mass following transection of the cranial cruciate ligament and immediate stifle stabilization, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 155, 1999.
35. Sikes R: Postoperative management of the neurosurgical patient, *Prob Vet Med* 3:467-477, 1989.
36. Olby N: Treatment of acute spinal cord injuries, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 145, 2004.
37. Olby N: Treatment of lumboscacral disease and other chronic spinal problems, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 149, 2004.
38. Sherman J: Physical rehabilitation for neurological patients: general approach and techniques, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 153, 2004.
39. McCauley L: Disc disease and paralysis, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 217, 2002.
40. McGonagle L: Peripheral nerve injuries, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 263, 2002.
41. Bockstahler B et al: Benefits of physical therapy for dogs undergoing decompressive spinal cord surgery, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 193, 2002.
42. Looney A: Rehabilitation considerations in critically injured, paralyzed, and immobilized patients, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 167, 2002.
43. Sharon, L: Physical therapy in a case of polyradiculoneuritis, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 121, 2002.
44. Roth R: Rehabilitation of the canine with intervertebral disc disease, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 117, 2002.
45. Thomas WB: Canine neurological conditions, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 87, 2002.
46. Bockstahler B et al: Benefits of physical therapy for dogs undergoing decompressive spinal cord surgery, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 193, 2002.
47. Albrecht WL: Combination of veterinary medicine, physical therapy and orthotics in a brachial plexus nerve degeneration: case study, *Proceedings of The First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 145, 1999.
48. Alexander CS: Rehabilitation of the paretic patient, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* August 10-14, Knoxville, Tenn, 119, 2002.

49. Nwadike BS: Rehabilitation of fracture patients, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 141, 2004.
50. Millis DL: Managing chronic osteoarthritis using physical rehabilitation, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 127, 2004.
51. Taylor RA: Canine hip dysplasia—Treatment and rehabilitation, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 131, 2004.
52. Marcellin-Little DJ: The postoperative hip, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* August 10-14, Knoxville, Tenn, 233, 2002.
53. Adamson CP, Taylor RA: Preliminary functional outcomes of extracorporeal shockwave therapy on ten dogs with various orthopedic conditions, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 195, 2002.
54. Finster ST, Marcellin-Little DJ: Risk analysis of goniometric changes present in Labrador retrievers with developmental orthopedic diseases, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 185, 2002.
55. Marcellin-Little DJ: Rehabilitation of fractures, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 154, 2002.
56. Hamilton SA: Rehabilitation of osteoarthritis in a dog, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 127, 2002.
57. Roth R: Rehabilitation of the canine with intervertebral disc disease, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 117, 2002.
58. Johnston KD et al: The effects of TENS on osteoarthritic pain in the stifle of dogs, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 199, 2002.
59. Millis DL, Levine D: Arthritis management, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 135, 2002.
60. Alexander C: Experiences with physical therapy in various orthopaedic disorders, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 171, 1999.
61. Dee JF: Medical management of osteoarthritis in senior dogs, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 83, 1999.
62. Payne JT: General management considerations for the trauma patient, *Vet Clin North Am Small Anim Pract* 25:1015-1029, 1995.
63. Steiss JE: Physical therapy/rehabilitation of injuries in sporting dogs, *Proceedings of the Sixth Annual Sports Medicine Program Symposium,* Jan 30, Auburn, Ala, 59, 1999.
64. Manning AM et al: Physical therapy for critically ill veterinary patients. I. Chest physical therapy, *Compend Cont Educ Pract Vet* 19:675-689, 1997.
65. Manning AM et al: Physical therapy for critically ill veterinary patients. II. The musculoskeletal system, *Compend Cont Educ Pract Vet* 19:803-807, 1997.
66. Clark GN: Traumatic injury patterns in canine athletes, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 143, 1999.
67. Hansen B: Comprehensive management of acute pain, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 73, 2004.
68. Lascelles DX: Comprehensive management of chronic animal pain—the rationale for multimodal therapy for chronic pain, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 79, 2004.
69. Roth R: Animal palliative and hospice care rehab in reverse, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 131, 2002.
70. Frick AL: Pain management without the pill, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 129, 2002.
71. Downer AH: *Physical therapy for animals: selected techniques*, Springfield, Ill, 1978, Charles C Thomas.
72. Bloomberg MS et al: *Canine sports medicine and surgery*, Philadelphia, 1998, WB Saunders.
73. Blythe LL et al: *Care of the racing and retired greyhound*, Abilene, Kan, 2007, American Greyhound Council.
74. Blythe LL et al: *Care of the racing greyhound*, Abilene, Kan, 1994, American Greyhound Council.
75. Millis DL, Levine D, Taylor RA: *Canine rehabilitation and physical therapy*, St Louis, 2004, WB Saunders.
76. Bockstahler B et al: *Essential facts of physiotherapy in dogs and cats: rehabilitation and pain management*, Germany, 2004, BE Vet Verlag.
77. McGowan C et al: *Animal physiotherapy: assessment, treatment and rehabilitation of animals*, Oxford, 2007, Blackwell.
78. Marcellin-Little DJ: Assistive ambulation devices, *Proceedings of the Third International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, NC State, NC, 277, 2004.
79. Hamilton S: Orthotics, slings, and carts, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 241, 2002.
80. Hamilton S: Aquatic exercise for the small animal patient, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 163, 2002.
81. Penick T: Aquatics in canine rehabilitation, *Proceedings of the Second International Symposium on Rehabilitation and*

Physical Therapy in Veterinary Medicine, Aug 10-14, Knoxville, Tenn, 159, 2002.

82. Saunders DG: Electrical stimulation, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 55, 2002.
83. Steiss JE: Therapeutic ultrasound, *Proceedings of the Second International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 10-14, Knoxville, Tenn, 49, 2002.
84. Steiss JE: Therapeutic ultrasound: theory and application principles, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 43, 1999.
85. Steiss JE: Magnetic field therapy: theory and application principles, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 51, 1999.
86. Levine D, Millis DL: Electrical stimulation, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 55, 1999.
87. Dyson M: Low intensity laser therapy: theory and application principles, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 61, 1999.
88. Schoen AM: Veterinary acupuncture: theory and principles of application in rehabilitation, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 65, 1999.
89. Dumbleton CD: Electrotherapy and the greyhound, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 109, 1999.
90. Clark GN: Orthotics, prosthetics, and ambulatory carts: use of supportive devices in canine patients, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 141, 1999.
91. Glennerster M: Hydrotherapy and the racing greyhound, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 147, 1999.
92. le Jeune R: The Winks Greene machine—e-stim for animals, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 153, 1999.
93. Levine D et al: Thermal effects of 3 MHz ultrasound treatments in dogs, *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 161, 1999.
94. Millis DL et al: Variables affecting thigh circumference measurements in dogs. *Proceedings of the First International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Aug 7-11, Corvallis, Ore, 157, 1999.
95. Steiss JE, Adams CC: Effect of coat on rate of temperature increase in muscle during ultrasound treatment of dogs, *Am J Vet Res* 60:76-80, 1999.
96. Levine D et al: Effects of 3.3 MHz ultrasound on caudal thigh muscle temperature in dogs, *Vet Surg* 30:170-174, 2001.
97. Millis DL, Levine D: The role of exercise and physical modalities in the treatment of osteoarthritis, *Vet Clin North Am Small Animal Pract* 27:913-930, 1997.

2 Regulatory and Practice Issues for the Veterinary and Physical Therapy Professions

David Levine and Darryl Millis

Definition of Physical Therapy

The American Physical Therapy Association (APTA) has developed a model definition of *physical therapy.* In this model definition, physical therapy includes examining and evaluating patients with impairments, functional limitations, disability, and other health-related conditions to determine a diagnosis, prognosis, and intervention. Some examples of areas that may be examined include aerobic capacity, arousal, cognition, assistive and supportive devices, barriers, ergonomics, gait, balance, pain, posture, prosthetic requirements, and range of motion.[1]

Physical therapists also alleviate impairments and functional limitations by designing, implementing, and modifying therapeutic interventions. Some examples of these activities include therapeutic exercise, functional training, manual therapy techniques, electrotherapeutic modalities, and patient-related instruction. Physical therapists are also involved with helping to prevent injury, impairments, functional limitations, and disability, and with the promotion and maintenance of fitness, health, and quality of life in all age populations. Physical therapists accomplish these tasks in a variety of situations, including patient consultation, education, and research.

Definition of Veterinary Medicine

The practice of veterinary medicine is described in the directory of the American Veterinary Medical Association (AVMA) in a section titled The Model Practice Act (Act).[2] The Act has been periodically updated and revised, and in the current version (2012, Section 2.16), the practice of veterinary medicine means:

a. To diagnose, prognose, treat, correct, change, alleviate, or prevent animal disease, illness, pain, deformity, defect, injury, or other physical, dental, or mental conditions by any method or mode; including the:
   i. performance of any medical or surgical procedure, or
   ii. prescription, dispensing, administration, or application of any drug, medicine, biologic, apparatus, anesthetic, or other therapeutic or diagnostic substance, or
   iii. use of complementary, alternative, and integrative therapies, or
   iv. use of any procedure for reproductive management, including but not limited to the diagnosis or treatment of pregnancy, fertility, sterility, or infertility, or
   v. determination of the health, fitness, or soundness of an animal, or
   vi. rendering of advice or recommendation by any means including telephonic and other electronic communications with regard to any of the above.
b. To represent, directly or indirectly, publicly or privately, an ability and willingness to do an act described in subsection 16(a).
c. To use any title, words, abbreviation, or letters in a manner or under circumstances that induce the belief that the person using them is qualified to do any act described in subsection 16(a).

This updated Model Veterinary Practice Act includes the use of complementary, alternative, and integrative therapies. Until 2001 these had been listed under Guidelines for Alternative and Complementary Veterinary Medicine, which is now referred to as "Complementary, Alternative, and Integrative Therapies." These guidelines describe the potential use and application of these therapies for veterinary patients.[3] The Model Veterinary Practice Act has included these forms of treatment to help protect the public, by placing the definitions under the Practice of Veterinary Medicine. Approximately 20 states have followed the AVMA Model Practice Act and included complementary and alternative medicine in their state practice act definition of veterinary medicine, while another 20 have enacted exemptions for certain therapies, generally requiring veterinary supervision or referral.

Complementary, Alternative, and Integrative Therapies

The AVMA Guidelines for Complementary and Alternative Veterinary Medicine address topics such as veterinary acupuncture and acutherapy, and acupressure; low-energy photon therapy; magnetic field therapy; veterinary chiropractic; veterinary physical medicine and rehabilitation; veterinary homeopathy; and nutraceutical medicine.[3] The guidelines state:

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The AVMA believes that all veterinary medicine, including CAVM, should be held to the same standards. Claims for safety and effectiveness ultimately should be proven by the scientific method. Circumstances commonly require that veterinarians extrapolate information when formulating a course of therapy. Veterinarians should exercise caution in such circumstances. Practices and philosophies that are ineffective or unsafe should be discarded.

Furthermore,

"The AVMA believes veterinarians should ensure that they have the requisite skills and knowledge for any treatment modality they may consider using."

Recommendations for patient care state that:

> "The foremost objective in veterinary medicine is patient welfare. Ideally, sound veterinary medicine is effective, safe, proven, and holistic in that it considers all aspects of the animal patient in the context of its environment.
>
> Diagnosis should be based on sound, accepted principles of veterinary medicine. Proven treatment methods should be discussed with the owner or authorized agent when presenting the treatment options available. Owner consent should be obtained prior to initiating any treatment, including CAVM. Clients usually choose a medical course of action on the advice of their veterinarian. Recommendations for effective and safe care should be based on available scientific knowledge and the medical judgment of the veterinarian."

The veterinarian also undertakes responsibilities, including:

> "These guidelines support the requisite interaction described in the definition of the veterinarian-client-patient relationship.[1] Accordingly, a veterinarian should examine an animal and establish a preliminary diagnosis before any treatment is initiated.
>
> The quality of studies and reports pertaining to CAVM varies; therefore, it is incumbent on a veterinarian to critically evaluate the literature and other sources of information. Veterinarians and organizations providing or promoting CAVM are encouraged to join with the AVMA in advocating sound research necessary to establish proof of safety and efficacy.
>
> Medical records should meet statutory requirements. Information should be clear and complete. Records should contain documentation of client communications and owner consent.
>
> In general, veterinarians should not use treatments that conflict with state or federal regulations. Veterinarians should be aware that animal nutritional supplements and botanicals typically are not subject to premarketing evaluation by the FDA for purity, safety, or efficacy and may contain active pharmacologic agents or unknown substances. Manufacturers of veterinary devices may not be required to obtain premarketing approval by the FDA for assurance of safety or efficacy. Data establishing the efficacy and safety of such products and devices should ultimately be demonstrated. To assure the safety of the food supply, veterinarians should be judicious in the use of products or devices for the treatment of food-producing animals.
>
> If a human health hazard is anticipated in the course of a disease or as a result of therapy, it should be made known to the client."

Veterinary Physical Rehabilitation

Veterinary physical rehabilitation is the use of noninvasive techniques, excluding veterinary chiropractic, for the rehabilitation of injuries in nonhuman animals. Veterinary physical rehabilitation performed by nonveterinarians should be limited to the use of stretching; massage therapy; stimulation by use of low-level lasers, electrical sources, magnetic fields, and ultrasound; rehabilitative exercises; hydrotherapy; and applications of heat and cold. Veterinary physical rehabilitation should be performed by a licensed veterinarian or, where in accordance with state practice acts, by a licensed, certified, or registered veterinary or animal health technician educated in veterinary physical rehabilitation or a licensed physical therapist educated in nonhuman animal anatomy and physiology. Veterinary physical rehabilitation performed by a nonveterinarian should be performed under the supervision of, or referral by, a licensed veterinarian who is providing concurrent care.

Massage Therapy

Massage therapy is a technique in which the therapist uses only his or her hands and body to massage soft tissues. Massage therapy on nonhuman animals should be performed by a licensed veterinarian with education in massage therapy or, where in accordance with state veterinary practice acts, by a graduate of an accredited massage school who has been educated in nonhuman animal massage therapy. When performed by a nonveterinarian, massage therapy should be performed under the supervision of, or referral by, a licensed veterinarian who is providing concurrent care.[3]

The development of the AVMA Guidelines for Complementary and Alternative Veterinary Medicine is a recognition by the veterinary profession that other health care professionals have information and knowledge that may benefit veterinary patients. Veterinarians should not hesitate to seek the help, advice, and expertise of others to improve the care and management of their patients. When this expertise, assistance, and advice is sought, however, the responsibility for the care, diagnosis, treatment, and management of the patient remains with the attending veterinarian, and a veterinarian–client–patient relationship (VCPR) will be established.

The Veterinarian–Client–Patient Relationship

The concept of the VCPR was introduced by the AVMA to clearly describe the relationship of the veterinarian to clients (the owners) and patients served by the profession. This VCPR is the basis for professional interactions and has become part of a variety of official AVMA documents, including the Principles of Veterinary Medical Ethics, the Model Practice Act, and the Guidelines for Veterinary Prescription Drugs. The Guidelines for Complementary and Alternative Veterinary Medicine also refer to the VCPR, and, because these modalities are considered to be part of the practice of veterinary medicine, they should be offered only in the context of a valid VCPR.

As stated in the Principles of Veterinary Medical Ethics,[4] the VCPR is the basis for interaction among veterinarians, owners, and patients. A VCPR exists when all of the following conditions have been met:

1. The veterinarian has assumed responsibility for making clinical judgments regarding the health of the animal or animals and the need for medical treatment, and the client has agreed to follow the veterinarian's instructions.
2. The veterinarian has sufficient knowledge of the animals to initiate at least a general or preliminary diagnosis of the medical condition of the animals. This means that the veterinarian has recently seen and is personally acquainted with the keeping and care of the animals by virtue of an examination, or by medically appropriate and timely visits to the premises where the animals are kept.
3. The veterinarian is readily available, or has arranged for emergency coverage, for follow-up evaluation in the event of adverse reactions or the failure of the treatment regimen.

The Principles additionally state that when a VCPR exists, veterinarians must maintain medical records, which should contain information on the diagnosis, care, and treatment of patients.[4]

The Principles also discuss the termination of the VCPR, which applies when professional services are no longer assumed by the veterinarian or no longer needed by the client. Veterinarians may terminate a VCPR under certain conditions, and they have an ethical obligation to use courtesy and tact in discharging this responsibility. Guidelines for terminating the VCPR are as follows:

1. If there is no ongoing medical condition, veterinarians may terminate a VCPR by notifying the client that they no longer wish to serve that patient and client.
2. If there is an ongoing medical or surgical condition, the patient should be referred to another veterinarian for diagnosis, care, and treatment. The former attending veterinarian should continue to provide care, as needed, during the transition.

The Principles state that clients may terminate the VCPR at any time.[4]

Guidelines for Referrals

The expansion and growth of knowledge, the emergence of specialization, and the use of the services and expertise of other health care professionals have necessitated the increased use of referrals in veterinary medicine. Referrals are to be encouraged among veterinarians, and the Guidelines for Referrals[5] approved by the AVMA in 1990 and revised in 2009 should be followed when referrals are used. These referral guidelines are summarized and paraphrased as follows:

Definitions

The referring veterinarian is the veterinarian who was in charge of the patient at the time of the referral. The receiving veterinarian or the referral veterinarian is the veterinarian to whom a patient is sent either by referral or for consultation. A *consultation* is a deliberation between two or more veterinarians concerning the diagnosis of a disease and the proper management of the case. A *referral* is the transfer of responsibility of diagnosis, care, and treatment from the referring veterinarian to the receiving veterinarian, and a new VCPR is established with the receiving veterinarian.

In these descriptions and definitions, the assumption is made that referrals and consultations may occur only between veterinarians. The Guidelines for Referrals do not address a situation in which the individual who is referring or receiving the patient is not a veterinarian. If these Guidelines for Referrals apply only to veterinarians, then other health care professionals may not actually accept or recommend referrals without the involvement of a veterinarian and the establishment of a valid VCPR. However, when a treatment modality such as physical therapy is required, a referral directly to a therapist may be appropriate. In that instance, specific written instructions and orders must accompany the referral. Because of the necessity of a valid VCPR, other health care professionals who might be involved in the diagnosis, care, and treatment of veterinary patients must work closely with veterinarians so a valid VCPR will be established and maintained.

Method of Referral

When a referral is being considered, communication among veterinarians and other health care professionals is essential. Communications may occur by letter, telephone, direct contact, or other means, and the most appropriate method of communication should be determined by the individuals involved. The referring veterinarian should provide the receiving veterinarian with all the appropriate information pertinent to the case before or at the time of the first contact with the patient or the owner. When the referred patient has been examined and a diagnosis has been established, the referring veterinarian should be promptly informed of those findings. Information provided should include diagnosis, proposed care, treatment plans, and other recommendations. If the patient undergoes a prolonged treatment or hospitalization, then immediately on discharge of the patient, the referring veterinarian should receive a detailed and complete report, preferably written, and should be advised as to continuing care of the patient or termination of the case. Each veterinarian involved in the case is entitled to collect fees for service, care, and treatments for professional services; fee splitting is not allowed.[4]

All licensed veterinarians may receive referrals, and referring veterinarians may refer to whomever they believe appropriate for the situation. Referrals should occur in a timely manner, and veterinarians who solicit or encourage referrals should abide by the AVMA Principles of Veterinary Medical Ethics. The receiving veterinarian should provide only services or treatments relative to the referred condition and should consult the referring veterinarian if other services or treatments are needed.

When referrals are made to other health care providers who may not be licensed veterinarians, the same standards and guidelines must apply. Other health care professionals (nonveterinarians) who are involved with the care and treatment of veterinary patients cannot assume a valid VCPR, and therefore the ultimate responsibility for diagnosis, care, and treatment of the patient remains with the referring veterinarian, as a VCPR must be in effect at all times.

State Practice Acts: Veterinary Medicine and Physical Rehabilitation

Although interest in and support for the unique practice of veterinary physical rehabilitation is growing nationally, each state individually regulates the standards for both veterinary medicine and physical therapy through the respective practice acts for each profession. Both the APTA and the AVMA have adopted positions and policies that support animal physical rehabilitation.[3,6]

These organizations are not the regulatory bodies for the disciplines of physical therapy or veterinary medicine. The practice of veterinary medicine and that of physical therapy are regulated by respective practice acts for each state, and it is the responsibility of individuals who practice veterinary physical therapy to understand the legal issues related to both the veterinary and physical therapy practice acts of their respective states before pursuing this specialized discipline. Individuals practicing veterinary physical rehabilitation who fail to comply with the rules and regulations of the respective state practice acts could be practicing both veterinary medicine and physical therapy without a license and might be subject to investigation, warning, disciplinary action, or criminal prosecution. A useful resource is this list entitled "State-by-State Assessment of Physical Therapy and Veterinary Practice Acts."

http://www.orthopt.org

This resource also lists people in the state to contact.

Veterinary Medicine

The veterinary practice acts are often vague when referring to the practice of veterinary physical rehabilitation. One exception is the Idaho Veterinary Practice Act, which states that "The requirement to have a license to practice veterinary medicine does not prohibit an allied health professional actively licensed and in good standing in any state from participating in a medical procedure involving an animal, provided that such participation is in his licensed field of medicine and under the indirect supervision of an actively licensed veterinarian."[7] "An allied health professional" means a person holding a current active license, in good standing, in any state to practice one of the healing arts including, but not limited to, medicine, dentistry, osteopathy, chiropractic, acupuncture and podiatry." Indirect supervision means "the supervising veterinarian is not on the premises but is available for immediate contact by telephone, radio or other means, has given either written or oral instructions for treatment of the animal patient, the animal has been examined by the supervising veterinarian as acceptable veterinary medical practice requires, and the animal, if previously anesthetized, has recovered to the point of being conscious and sternal."[7] The supervising veterinarian shall "Have examined the animal patient prior to the delegation of any animal health care task to a certified veterinary technician, temporary certification holder, or assistant. The examination of the animal patient shall be conducted at such times as acceptable veterinary medical practice dictates, consistent with the particular delegated animal health care task."

In the state of Pennsylvania, both the State Board of Veterinary Medicine and the State Board of Physical Therapy are regulated by the same administration.[8] This arrangement allows for a clear understanding of the regulatory issues affecting both veterinarians and physical therapists when members of either profession are interested in veterinary physical rehabilitation. However, individuals practicing their respective professions in that state must be licensed by the regulatory board of their respective profession. The State Board of Veterinary Medicine states that "Veterinarians may seek, through consultation, the assistance of other licensed professionals, including chiropractors, dentists, dental hygienists and physical therapists, when it appears that chiropractic, dental, dental hygiene or physical therapy procedures will enhance the quality of veterinary care. Chiropractic, dental, dental hygiene and physical therapy procedures shall only be performed upon animals by chiropractors, dentists, dental hygienists and physical therapists in conjunction with the practice of veterinary medicine and under the direct supervision of a veterinarian, subject to a limitation provided by law or regulation."

Physical Therapy

The physical therapy practice acts of many states include variable wording for authorizing and sanctioning the application of physical therapy techniques and procedures to animals and to the direct access of owners to physical therapy services. Currently, many states use the words *human* or *human being,* some states use the word *person,* and others use the word *individual* in their physical therapy practice acts (Box 2-1). This wording may be a problem for physical therapists who wish to practice veterinary physical therapy in those states, because the state practice acts may actually prohibit physical therapists from providing care to animals. Obtaining continuing education units or credits for courses in animal rehabilitation may also be problematic if the state physical therapy board does not accept this area of practice as legal. Direct access to physical therapy by human patients is an intriguing issue, in that most states permit direct access to either "physical therapy evaluation" or "physical therapy evaluation and treatment" without a physician referral. In the states that allow direct access of human patients to physical therapy, a physical therapist does not have to act with a physician to initiate physical therapy modalities; however, such direct access is not allowed for veterinary patients because of the definitions of veterinary practice and the necessity for a valid VCPR.[2] The physical therapy state practice acts are readily available online.[9]

Because of the growing interest in animal rehabilitation and physical therapy, the Animal Physical Therapist Special Interest Group (APTSIG) within the Orthopedic Section of the APTA has collected information regarding the state practice acts and animal physical therapy. This information is intended to identify which state practice acts discuss animal rehabilitation and physical therapy, and which state practice acts include the words *humans, people,* or *persons,* which might limit the practice of animal rehabilitation and physical therapy in those states. The APTSIG is also collecting information regarding the veterinary practice acts in all states.

In many states the terms *physical therapy* and *physical therapist* are "protected terms," and therefore only licensed practitioners of physical therapy in those jurisdictions may lawfully represent themselves as physical therapists or state that they provide physical therapy services. Veterinarians or veterinary technicians practicing physical therapy on animals in these jurisdictions should state that they provide physical rehabilitation, or rehabilitation, to avoid potential conflicts, unless they employ a physical therapist or physical therapist assistant (PTA) in their practice.

The physical therapy state practice act of New Mexico is a physical therapy practice act that specifically addresses veterinary physical therapy, and in that act a doctor of veterinary medicine is described as a "primary health care provider."[10] This statement occurs in the section of the practice act pertaining to direct care (access) requirements for physical therapists. In this practice act a "primary health care provider" is defined as

> a health care professional who is licensed in the U.S. and provides the first level of basic or general health care for individual's health needs including diagnostic and treatment services and includes but is not limited to a physician (M.D., D.O., D.P.M.), doctor of veterinary medicine (D.V.M.), doctor of chiropractic (D.C.), doctor of dental surgery (D.D.S.), doctor of oriental medicine (D.O.M.), certified nurse practitioner (C.N.P.), certified nurse-midwife (C.N.M.), licensed midwife (L.M.), and physician assistant (P.A.) practicing under the auspices of one of the providers listed herein.

Box 2-1 Physical Therapy State Practice Acts

Human or Human Beings

Alabama
Alaska
Arkansas
Colorado
Florida
Georgia
Hawaii
Idaho
Iowa
Maine
Missouri
Montana
New York
Oregon
Rhode Island
South Carolina
Utah
Washington, DC
Wyoming

Persons

Arizona
California
Connecticut
Delaware
Illinois
Minnesota
Nebraska
Nevada
North Carolina
Ohio
Pennsylvania
Texas*
Vermont
Washington
West Virginia

Patients

Louisiana
Massachusetts
New Mexico
South Dakota

Individuals

Kansas
Kentucky
Maryland
Michigan†
Oklahoma
Tennessee
Virginia

No Specification

Indiana
Mississippi
New Hampshire
New Jersey
North Dakota
Wisconsin

*Part C states, "treatment, consultative, educational, or advisory services to reduce the incidence or severity of disability or pain to enable, train, or retrain a person to perform the independent skills and activities of daily living."
†Defines *individual* as "natural person."

The state physical therapy practice act of New Mexico further defines direct care requirements as follows:

> A physical therapist shall not accept a patient for treatment without an existing medical diagnosis for the specific medical or physical problem made by a licensed primary care provider, except for those children participating in special education programs … and for acute care within the scope of the practice of physical therapy.

The act continues to define direct care requirements with the following statement:

> When physical therapy services are commenced under the same diagnosis, such diagnosis and plan of treatment must be communicated to the patient's primary health-care provider at intervals of at least once every sixty (60) days, unless otherwise indicated by the primary care provider. Such communication will be deemed complete as noted in the patient's medical record by the physical therapist.[10]

Fundamentally, the New Mexico state physical therapy practice act allows a physical therapist to obtain referrals from a veterinarian and to have direct access to veterinary physical therapy patients who are in need of acute care physical therapy, as long as communication occurs between the veterinarian and the physical therapist within 60 days of initiating treatment and regularly every 60 days thereafter. Interestingly, the New Mexico veterinary practice act also states that because a physical therapist must work under direct supervision of a veterinarian, the fee for physical therapy services must be paid to the licensed veterinarian or licensed veterinary facility.

This is an example in which the practice acts for veterinary medicine and physical therapy are apparently in conflict. Although some state physical therapy acts allow the treatment of animals by physical therapists, the veterinary practice acts in those states may not allow the same degree of freedom; therefore, it is important for the physical therapist and the veterinarian to understand both the physical therapy and the veterinary practice acts in the states where they are working. Although the veterinary practice acts in most states do not address the subjects of who may perform veterinary physical rehabilitation and direct access to patients by physical therapists, the veterinary profession has been careful to monitor who may practice veterinary medicine. State veterinary boards have been active in restraining the practice of veterinary medicine without a license, and the AVMA, through various councils and committees, has promoted the importance of the VCPR. These activities ensure that a veterinarian is the primary health care provider who manages the diagnosis, care, and treatment of the veterinary patients. In addition, the association with other professions to enhance and improve the quality of veterinary care has been encouraged by the veterinary profession, and these relationships allow for a role of other health care professionals such as physical therapists in the improvement of such care.

Collaborative Relationships between the Professions

In 1993 the APTA House of Delegates approved and adopted a position statement concerning collaborative relationships between physical therapists and veterinarians. This position statement reads as follows:

> The American Physical Therapy Association endorses the position that physical therapists may establish collaborative, collegial relationships with veterinarians for purposes of providing physical therapy services or consultation. Physical therapists are the provider of choice for the provision of physical services regardless of the client. A collegial relationship is advantageous to the client.[6]

On the national level, collegial relationships have been established between physical therapists and veterinarians for several years with a national liaison between the AVMA and the APTA. International meetings, such as the International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine, which has been held every 2 years since 1999, is another example of the growing collaboration between the two professions. The International Association of Veterinary Rehabilitation and Physical Therapy (www.iavrpt.org) sponsors this meeting. Other international collaborative organizations have been formed to foster the growth and document the status of veterinary physical rehabilitation in other nations. In some countries such as the United Kingdom and the Netherlands, physical therapists have been performing veterinary physical therapy in collaboration with veterinarians for many years, and the experiences of practitioners in these countries are being shared and exchanged.

There are many ways for physical therapists and veterinarians to work together. Veterinary colleges could employ physical therapists to provide appropriate care and therapy to patients requiring rehabilitation. Faculty from universities with colleges of veterinary medicine and physical therapy departments could work together to provide clinical service, teaching, and research activities that would benefit students and faculty members from both disciplines. Veterinary hospitals and clinics could employ physical therapists with either full-time or part-time appointments, or these hospitals could consult with physical therapists as needed. In jurisdictions where permitted such as in Colorado, physical therapists could establish veterinary rehabilitation therapy practices and solicit referrals from veterinarians in accordance with the state practice acts. Other venues where physical therapists might work include zoos; wildlife rehabilitation centers and parks; and canine and equine performance centers, exhibitions, and rodeos.

When collaborative relationships are initiated between veterinarians and physical therapists, the veterinarian must first establish a VCPR and then develop a care and treatment plan after a diagnosis is confirmed. This information must be communicated to the physical therapist, preferably in writing, before the physical rehabilitation is initiated. Such communications are analogous to writing "doctor's orders" in a medical record. The physical therapist may wish to perform an independent evaluation and confirm a course of treatment in conjunction with the veterinarian. If opinions differ on the appropriate diagnosis and course of treatment, these differences should be negotiated in a professional manner. After the treatment plan is initiated, contacts and communications should continue between the physical therapist and the veterinarian regarding the progress and response of the patient throughout the course of therapy. As in human medicine, in which the physician is ultimately responsible for the patient, the veterinarian must be ultimately responsible for the care and treatment of the patient, because a VCPR must be maintained.

Risk Management for Physical Therapy and Veterinary Professionals

As physical therapists are finding new opportunities for professional services, including veterinary medicine, participation in these situations opens new questions about conduct and professional practice. There is no doubt that physical therapists have knowledge and skills that can benefit veterinary patients, but the legal and ethical issues regarding these collaborative relationships sometimes are unclear. Both the AVMA and the APTA have been supportive of these collaborative relationships; however, many states do not recognize these activities as falling within the scope of physical therapy. For this reason, physical therapists who choose to practice with animals in any professional capacity should be certain that they are acting in compliance with the physical therapy and veterinary medical practice acts in the states where they are working.

As stated earlier, the APTSIG has established a liaison program to assist members with an interest in rehabilitation for animals. The goal of this program is to obtain copies of the various state veterinary medical and physical therapy practice acts and to produce a resource directory of individuals interested in veterinary rehabilitation. A national liaison coordinator for this program has been identified. While this liaison program is helpful, interested individuals should contact appropriate licensing boards for clarification of questions.

See Box 2-2 for guidelines that are recommended when physical therapy practitioners are interested in practicing veterinary rehabilitation.

Veterinary practice acts in most jurisdictions are quite clear on the issues of the definition of veterinary practice, access to veterinary patients, and who may practice veterinary medicine. In all cases, the establishment of a VCPR is mandatory, and this rule necessitates that a physical therapist who is involved with an animal patient work closely with the veterinarian who has established the VCPR with that owner and patient. Otherwise, a physical therapist practicing on that animal could be in violation of a state veterinary practice act. When a VCPR has been established and followed, any potential legal issues should be minimized.

See Box 2-3 for guidelines that are recommended when veterinary practitioners are interested in practicing veterinary physical rehabilitation.

Box 2-2 Guidelines for Physical Therapists on Practicing Veterinary Physical Rehabilitation

1. If the physical therapy practice act includes language that limits the practice to humans by using terminology such as *humans, people,* or *persons,* then practicing on animals may be in violation of that specific practice act. If the term *patient* is used, the language of that practice act may be permissive, and the physical therapist should check with the state licensing board about the exact wording and the current interpretation (see Box 2-1).
2. After reviewing the state physical therapy practice act, if the wording is still unclear, the physical therapy practitioner should contact the state licensing board for written clarification or consult with an attorney for a legal opinion.
3. If the activities of interest are clarified in the state practice act (and are within the scope of physical therapy practice as defined by the APTA), one should be able to obtain professional liability insurance for these activities. If activities are specifically excluded in the state practice act, it is unlikely that professional liability insurance will provide coverage or protection for potential litigation.
4. PTAs who are providing veterinary physical rehabilitation also need to be aware of their state practice act regarding their practicing under the supervision of a licensed physical therapist. This does not necessarily mean on-site supervision; however, it does mean that the physical therapist has seen the patient and decided on a plan of care before the PTA initiates treatment. If they are performing animal rehabilitation under the supervision of a veterinarian (but without supervision of aPT), but not calling it *physical therapy,* this may be permitted based on the veterinary practice act in that individual state.
5. If a state precludes either physical therapists or PTAs from practicing veterinary physical therapy based on the state's practice act, the physical therapists or PTAs still may be able to work under the direct supervision of a veterinarian, but should not call their practice *physical therapy;* rather they should term it *physical rehabilitation, animal rehabilitation, canine* or *equine rehabilitation,* or something similar. The choice to do this would still have the potential to be in violation of the state practice act and would need to be examined on an individual basis.

Box 2-3 Guidelines for Veterinary Practitioners on Practicing Veterinary Physical Rehabilitation

1. If the term *physical therapy* or *physical therapist* is a protected term in that state, the term *physical rehabilitation, rehabilitation,* or a similar term should be used.
2. If a veterinary technician, physical therapist, or PTA is to provide the veterinary physical rehabilitation, supervision by a veterinarian may be defined differently state by state, and the veterinarian should review this, especially if the individual is providing care off-site, such as in home visits.

Reimbursement and Remuneration for Services

The reimbursement for professional services is another issue that must be resolved when there are collaborative relationships between veterinarians and physical therapists. Both health care professionals are entitled to an appropriate fee, which should be based on the services rendered with consideration for education and training of the health care provider. If the physical therapist is an employee of a veterinarian, the fee for the animal rehabilitation may be included in the total charges for veterinary services. Because the physical therapist is an employee, that individual would be paid directly by the veterinary clinic where employment occurs and services are rendered.

If the physical therapist is self-employed or an independent contractor, a veterinarian can arrange a referral or consultation directly with a physical therapist; however, a VCPR must originate and be maintained with the attending veterinarian. In that instance, a separate or direct billing arrangement for veterinary physical rehabilitation services could be appropriate if all involved parties are in agreement.

If the physical therapist is an employee of another health care provider, then that provider could charge the veterinarian or the owner directly, depending on the financial arrangements negotiated at the onset of the treatment plan. The attending veterinarian is ultimately responsible for the diagnosis, care, and treatment of the patient and must write instructions or orders to the health care provider and physical therapist affiliated with that provider. The physical therapist is an employee of the health care provider and is paid by that entity.

The physical therapist is entitled to a reasonable fee for services rendered. If a veterinarian refers a patient to another veterinarian, physical therapist, or health care provider, the original veterinarian is not allowed to accept part of the professional fee for the referral of the patient and owner. Such an arrangement is called *fee splitting* or a *kickback;* it is unethical and is in violation of the Principles of Veterinary Medical Ethics.[4]

In conclusion, the affiliations and collaboration between physical therapists and veterinarians will continue to grow with benefits for patients, owners, veterinarians, and physical therapists. With evolution and additional collaborative experiences in the discipline of animal physical rehabilitation, the specific roles of physical therapists and veterinarians will be clarified as well. Communication and cooperation between the professions of veterinary medicine and physical therapy, the APTA and AVMA, and the respective licensing boards of each state are necessary to successfully integrate physical therapy into the practice of veterinary medicine.

REFERENCES

1. Guide to physical therapy practice, ed 2, *Phys Ther* 81:21-22, 2001.
2. *Model Practice Act*, Schaumburg, Ill, 2012, American Veterinary Medical Association.
3. AVMA Guidelines for Complementary and Alternative Veterinary Medicine. https://www.avma.org/KB/Policies/Pages/AVMA-Guidelines-for-Complementary-and-Alternative-Veterinary-Medicine.aspx.
4. The Veterinarian-Client-Patient Relationship. https://www.avma.org/KB/Policies/Pages/Principles-of-Veterinary-Medical-Ethics-of-the-AVMA.aspx.
5. Attending, Consulting, and Referring. https://www.avma.org/KB/Policies/Pages/Principles-of-Veterinary-Medical-Ethics-of-the-AVMA.aspx.
6. Position on physical therapists in collaborative relationships with veterinarians, *American Physical Therapy Association House of Delegates* 06-93-20-36 (Program 32), 1993.
7. *Idaho Veterinary Practice Act*, Boise, Idaho, 2013, Idaho Board of Veterinary Medicine, Idaho Code 54-2103 (3), 54-2104 (2p) and Board Rules IDAPA 46.01.01.103.02.f.
8. *State Board of Veterinary Medicine and State Board of Physical Therapy, Professional and Vocational Standards*, Harrisburg, PA, 2013, Department of State, Pennsylvania Code.
9. www.utcaninerehab.com/practice-acts.asp.
10. State of New Mexico Physical Therapy Board, Regulation and Licensing Department: *Statutes and Regulations*, 1997, Title 16, Chapter 20, Part 10.

3 Conceptual Overview of Physical Therapy, Veterinary Medicine, and Canine Physical Rehabilitation

David Levine, Caroline P. Adamson, and Anna Bergh

The purpose of this chapter is to provide a conceptual framework outlining the general practice of canine physical rehabilitation and physical therapy. Background information regarding the professions of physical therapy and veterinary medicine is included to assist professionals from each discipline to understand the history, educational requirements, and current practice of each profession. Methods to effectively bridge the gap between the physical therapy and veterinary professions are addressed, with models for collaborative practice outlined.

Physical Therapy as a Profession

History of the Physical Therapy Profession in the United States

Physical therapy began in the United States in the early 1900s and focused on treatment of acute anterior poliomyelitis, which reached its peak during the first 2 decades of the twentieth century. At this time physical therapy was not a true occupation; however, the foundations for the profession were developed. Some of these early applications of physical therapy included exercise, massage, and certain physical agent modalities.[1] The need for physical rehabilitation during and immediately following World War I served to further enhance the emerging field of physical therapy. From its origins, physical therapy has focused on restoring maximal function to individuals with disabilities.

Formal training in physical therapy began around 1918 and was developed by cooperative efforts between the office of the Surgeon General and personnel in civilian institutions. Individuals who completed these training courses were given the title of *reconstruction aides,* the earliest title of physical therapists.[1] Many of these individuals worked in the military during this time. The first national organization was the American Women's Physical Therapeutic Association, which was founded in 1921. In 1922 this name was changed to the American Physiotherapy Association, and in 1947 to the American Physical Therapy Association (APTA).[1] Throughout the 1940s physical therapy continued to evolve and focused on treating patients who had contracted polio or those injured in World War II. This was a period of major growth in physical therapy, and because of the shortage of physical therapists, many more had to be trained during this time. From the 1940s to the present, physical therapy has gradually become a more autonomous and scientifically based profession. Physical therapy is an accepted medical intervention, and approximately 750,000 people are treated by physical therapists in the United States each day.[2]

The American Physical Therapy Association

The APTA is a national professional organization representing more than 71,000 physical therapists, physical therapist assistants, and students in the United States.[3] Membership in the APTA is not mandatory for physical therapists practicing in the United States, and currently more than 172,000 physical therapists are licensed in the United States.

Major responsibilities of the APTA are monitoring and improving physical therapy education, practice, and research and educating the general public about the role of physical therapy in health care. The APTA political action committee (PAC) is one of the largest health care PACs in the nation. It was formed to empower the physical therapist profession to be more involved in the determination of federal laws and policies.

Physical therapists practice in many settings, including hospitals, school systems, private practices, extended care facilities and nursing homes, home health agencies, academic institutions, research centers, and government agencies. The APTA has 19 specialty sections, which represent various areas in physical therapy. These include acute care aquatic physical therapy, cardiovascular and pulmonary, clinical electrophysiology and wound management, education, federal physical therapy, geriatrics, hand rehabilitation, health policy and administration, home health, neurology, oncology, orthopedics, pediatrics, private practice, research, sports physical therapy, and women's health. A physical therapist may become board certified by the American Board of Physical Therapy Specialties (Diplomate, ABPTS) in one of seven areas (cardiopulmonary, clinical electrophysiology, geriatrics, neurology, orthopedics, pediatrics, and sports physical therapy).

The orthopedic section of the APTA houses the animal physical therapist special interest group, which was formed in 1998 and has quickly grown to more than 400 members. The goals of this group are to:

- Promote physical therapy
- Share information
- Collaborate with other health professionals
- Develop educational programs
- Foster research
- Create guidelines for practice
- Encourage appropriate legislative changes
- Establish a national network
- Protect professional practice

Physical Therapist and Physical Therapist Assistant Education

The current entry-level physical therapy degree is a master's or doctoral degree. In the past, the entry-level degree was a bachelor's degree. The APTA supports changing the entry-level degree to the doctoral level by the year 2020. There are approximately 200 APTA-accredited physical therapy educational programs in the United States, which graduate roughly 5700 physical therapists each year. Every physical therapist must pass a national licensing examination. Additional requirements for practice vary from state to state.

Physical therapist assistants complete a 2-year associate's degree from an APTA-accredited program. This training prepares them to provide therapeutic interventions that have been delegated by their supervising physical therapist, but assistants cannot evaluate or prescribe treatment. Requirements for licensure vary from state to state, as do the continuing education requirements.

Educational backgrounds of both physical therapists and physical therapist assistants do not include any formal training in the rehabilitation of animals. Animal anatomy may be studied during undergraduate education, but most likely this will be anatomy of the cat. Application of physical therapy to animals is not included in standard curricula; however, a few physical therapy programs now offer elective coursework in this area.

Veterinary Medicine as a Profession

History of the Veterinary Profession

The first college of veterinary medicine was established at Lyon, France. The first college of veterinary medicine in the United States was established at Iowa State University in the 1800s. The initial emphasis in veterinary medicine was on agricultural production and livestock. Gradually the emphasis on the treatment of individual animals shifted to herd management. The emergence of companion animals as members of the family, combined with the shift in agricultural demand, has resulted in the development of small-animal and equine practice as the predominant emphasis in veterinary medicine. Furthermore, the explosion of knowledge in small-animal medicine and surgery, combined with the era of specialization, has resulted in improved health care for pets. One area that has been neglected until recently is the rehabilitation of animals with chronic ailments and following surgery. The goals of this new area are to increase function, improve the ultimate outcome of patients following major surgery, and enhance the quality of life.

The American Veterinary Medical Association

The national professional organization that represents the approximately 84,000 veterinarians in the United States is the American Veterinary Medical Association (AVMA).[4] The AVMA is responsible for evaluating and credentialing veterinary medical education, administering the national board examination for veterinarians, and overseeing the various specialty colleges (currently there are 21). The American College of Veterinary Sports Medicine and Rehabilitation, American College of Veterinary Surgeons; American College of Veterinary Internal Medicine, neurology subspecialty; American College of Veterinary Emergency and Critical Care; American Board of Veterinary Practitioners; American College of Veterinary Behaviorists; and American College of Veterinary Nutrition are specialty colleges or boards whose members are likely to treat patients that may benefit from rehabilitation. Examples of other related organizations include the American Academy of Veterinary Acupuncture and the American Veterinary Chiropractic Association.

Veterinarians undergo broad training that includes studying diseases of large, small, and exotic animals. Veterinarians undergo training in making medical diagnoses, with the aim to identify the pathophysiologic nature of the condition and its subsequent treatment. Currently, the American College of Veterinary Sports Medicine and Rehabilitation has received approval by the American Board of Veterinary of Specialties, which oversees specialties of the AVMA. Training in this area is relatively limited, although a number of colleges now offer courses or lectures in physical rehabilitation.

Veterinary and Veterinary Technician Education

Veterinarians usually have an average of 4 years of undergraduate education, followed by 4 years of professional curriculum at an AVMA-approved college. Currently there are 33 approved colleges of veterinary medicine in the United States and Canada graduating approximately 2100 veterinarians yearly.[4] In addition, there are now 13 non-Canadian foreign colleges of veterinary medicine that are approved by the AVMA. Veterinarians are required to pass national and, in some cases, state board examinations to practice. Some graduate veterinarians pursue additional training in the form of general internships and residencies in specialties. Specialty certification requires an internship or equivalent training, completion of a formal 2- to 3-year residency program, publication and research requirements, and successful completion of a certifying examination. There are currently 21 specialties recognized by the AVMA.

Veterinary technicians complete at least 2 years at an AVMA-accredited program and receive at least an associate's degree. There are currently nine distance learning programs in veterinary technology accredited or in the process of accreditation by the AVMA Committee on Veterinary Technician Education and Activities. In approximately 40 states and provinces, veterinary technicians are certified, registered, or licensed.[5] Candidates are tested for competency through an examination that may include oral, written, and practical portions. A state board of veterinary examiners or the appropriate state agency regulates this process. A national examination is available; however, requirements vary by individual states. Veterinary technician specialty organizations recognized by the North American Veterinary Technician Association include the Academy of Veterinary Emergency Critical Care Technicians, Academy of Veterinary Technician Anesthetists, Academy of Veterinary Surgical Technicians, Academy of Veterinary Dental Technicians, Academy of Internal Medicine for Veterinary Technicians, Academy of Equine Veterinary Nursing Technicians, Academy of Veterinary Nutrition Technicians, and the Academy of Veterinary Behavior Technicians.[6]

Veterinary and veterinary technician programs do not typically include any course work in physical rehabilitation. Although most veterinarians and veterinary technicians have a basic understanding of the rehabilitation process, they generally do not receive any formal training in this area. Application of physical rehabilitation to animals is not included in the standard curricula; however, a few veterinary programs now offer lectures or even elective courses in this area.

Continuing Education

Continuing education is essential to gain the knowledge in the reciprocal field that practitioners need to effectively perform canine rehabilitation and physical therapy. Some ways of accomplishing this are attending continuing education courses; self-guided study; attending meetings of the other profession; volunteering with members of the other profession; and attending related courses such as animal training, handling, and behavior.

A number of courses are offered in this area throughout the world. The First International Symposium on Rehabilitation and Physical Therapy, which was held at Oregon State University in 1999, helped to bring the professions of physical therapy and veterinary medicine together in a professional meeting. Research has been identified as critical to bridge the gap between professions and to establish the efficacy and credibility of physical therapy protocols and treatment. There are educational programs in the United States, the Netherlands, Japan, mainland Europe, Australia, and the United Kingdom.

Physical Therapy Evaluation and Intervention

Physical therapy encompasses a spectrum of services for humans, including evaluation, intervention, assessment, consultation, education, and research.[2] Evaluation includes taking a history; evaluating body systems to be certain that physical therapy is the appropriate medical intervention; and administering a variety of tests and measurements to determine a diagnosis, prognosis, and intervention. Physical therapists assess aerobic capacity and endurance; joint motion and integrity; muscle strength; arousal and cognition; need for assistive and adaptive devices; cranial nerve integrity; environmental barriers; body mechanics; gait; locomotion; balance; skin integrity; motor function; neuromotor development and sensory integration; orthotic, protective, and prosthetic devices; pain; posture; reflexes; circulation; and edema.[2]

The physical therapy diagnosis is both a process and label in which physical therapists perform an evaluation to provide a diagnose that identifies the effect a condition has on function.[2,7] Physical therapists proceed from the functional limitations that the animal presents and investigate the structures responsible for the impaired function. Based on the diagnosis, the physical therapist is able to specifically design individual treatment plans and evaluate their interventions

Physical therapists alleviate impairment and functional limitation by designing, implementing, and modifying therapeutic interventions that include the following[2]:

- Therapeutic exercise (range of motion [ROM], aerobic conditioning, gait training, aquatic therapy, muscle strengthening, balance, coordination, posture, motor control)
- Manual therapy techniques (joint mobilization and manipulation, massage, remodeling scar tissue)
- Wound management (dressings, topical agents, debridement, modalities, oxygen therapy)
- Airway clearance techniques (postural drainage, percussion, vibration, shaking)
- Orthotic and prosthetic intervention
- Electrotherapeutic modalities (electrical stimulation, laser)
- Thermal modalities (superficial heat and cold, and deep heat including ultrasound and diathermy)

Some of the common goals of physical therapy treatment are to decrease pain, improve muscle strength, retard atrophy, decrease swelling, decrease muscle spasm, increase the rate of tissue healing, remodel scar tissue, and improve function and independence in activities of daily living. Prevention of injury, impairment, functional limitation, and disability is also part of physical therapy. This includes promoting and maintaining fitness, health, and quality of life in all age populations.

Bridging the Gap between Physical Therapy and Veterinary Medicine

Traditional Physical Therapy and Veterinary Models of Practice

Physical therapists commonly practice by referral from a physician, although more than 30 states currently allow direct access to a physical therapist, meaning that a physical therapist practicing in these states may see a patient without physician referral. Additionally, some states allow direct access for evaluation only, meaning that after the evaluation is complete, the patient must still see a physician to allow treatment to begin. In this model, the physical therapist typically relates the findings to the physician to expedite the initiation of treatment. As discussed in Chapter 2, the veterinary practice acts in most states require the physical therapist to work under the direct supervision of a veterinarian (although there are some exceptions), thus precluding a physical therapist from direct access in the care of veterinary patients.

Veterinarians in general practice see a variety of patients with various conditions. In some cases, a referral to a specialist is made for treatment requiring advanced procedures and equipment. In most cases the referred patient is returned to the referring veterinarian for follow-up care and any general patient concerns. Veterinary technicians assist with the diagnostic and treatment procedures under the direction of the veterinarian.

Box 3-1 List of Individuals Contributing to Animal Rehabilitation

- Referring veterinarian
- Veterinary specialist (surgeon, neurologist, internist, emergency and critical care veterinarian)
- Rehabilitation veterinarian
- Physical therapist on staff or as a consultant
- Animal behaviorist
- Veterinary nutritionist
- Veterinary anesthesiologist
- Veterinary acupuncturist
- Veterinary chiropractor
- Veterinary technician
- Physical therapist assistant
- Support staff
- Orthotist/Prosthetist
- Biomedical engineer
- Owner

Team Approach to Rehabilitation

The individuals listed in Box 3-1 may be involved in a practice offering animal rehabilitation. The attending veterinarian is often the primary person responsible for the medical diagnosis and decisions regarding appropriate rehabilitative care. Depending on the injury and repair, specific recommendations are given to the person responsible for the rehabilitation. Precautions to therapy, especially exercises, must be clearly communicated. When veterinarians initiate physical rehabilitation in their practices, the initial sessions should be performed with all team members present, as differences in terminology may cause confusion and possibly result in injury. For example, performing ROM exercises may have different meaning to veterinarians, owners, and physical therapists. Communication and documentation between the team members are critical, and forms such as those illustrated later in this chapter provide an easy way to facilitate this.

The team approach involves a group evaluation or an assessment of the same patient by two or more clinicians within a short period. A core group of clinicians—for example, a veterinarian, physical therapist, and veterinary technician—is identified as the initial team. The initial team documents their findings, meets, and decides together the most effective plan for intervention. The core group may request additional evaluations by other experts on the team on an individual case basis. The appropriate clinicians are chosen to initiate the intervention plan. The team reviews the progress of the client periodically. One member of the team is responsible for follow-up after discharge.

This team approach allows for a variety of perspectives on health care issues. The team offers the client a detailed interdisciplinary assessment as well as the most cost-efficient method of providing services to resolve or treat the problem.

Collaborative Model for Practice

A proposed model for canine physical rehabilitation is outlined in Figure 3-1 and is based on a similar model for humans.

Documentation

Documentation should provide clear communication between all of the parties involved in the animal's care, including the veterinarian, the referring veterinarian (if applicable), the person responsible for the physical rehabilitation, and in some cases the owner. Because documentation is a critical form of communication, all patient files should be updated and documented at each physical rehabilitation treatment. Notes should be written in a clearly understandable, legible manner. An important characteristic of documentation is that the information should be easily obtained in a short period. A physical rehabilitation chart should give easy access to pertinent information such as functional status, ROM, treatment given, and patient

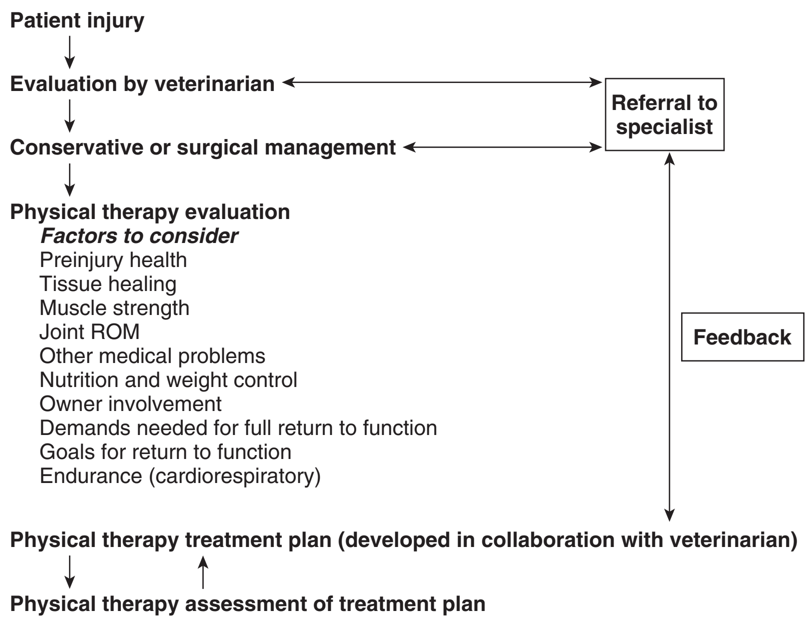


Figure 3-1 Proposed model for canine physical rehabilitation.

progress. Because terminology differs between professions, a glossary is included at the end of this book. An example of one difference is the directional term *anterior-posterior* used in human medicine versus *cranial-caudal* used in veterinary medicine. A treatment given once daily is written as *qd* in the human field, but *sid* in the veterinary profession (although qd is also used in veterinary medicine). Terms such as *closed kinetic chain exercises* (when the limb is in a weight-bearing position, such as in the weight-bearing or stance phase of walking) are not typically discussed in the veterinary field.

Documentation is also important from a legal perspective, as a record of the animal's condition and a clear documentation of the procedures performed. For example, consider an animal that has just completed a therapy session in which it demonstrated weight bearing on the affected extremity 75% of the time at a slow walk, and had a stifle ROM of 50 degrees flexion and 150 degrees extension, which was documented in the medical record. If this animal presents with non–weight-bearing lameness at the next treatment with a ROM of 60 degrees flexion and 130 degrees extension, that lends support to the idea that the animal had been injured not during therapy, but rather at home, in its cage, during a walk, or by some other means. Regardless of the mechanism of injury, the documentation may protect the therapist from accusations of fault, and also alert the therapist to contact the veterinarian. When an animal's status has changed, an evaluation by the veterinarian is necessary before therapy continues.

Another purpose of documentation is to verify the benefits of the treatment. The progress of the animal may be determined so that alterations can be made to the treatment protocol to further enhance recovery. Careful documentation and record keeping using objective assessment tools and standardized charts also provide a valid base for research. Conducting periodic chart audits can assist patient care in a variety of ways, including ensuring that all pertinent and necessary information is included, and that the treatment protocol is appropriate for the diagnosis.

The physical therapy chart should include the following key pieces of information:

1. A relevant medical history
2. The history of the present illness, including the attending veterinarian's referral information containing any precautions or contraindications for therapy, and treatment to be given
3. Any objective information such as ROM, function, limb girth measurements, and lameness scores
4. Primary problems to be addressed and goals for the treatment
5. The treatment itself in as much detail as possible
6. The response to treatment

Examples of documentation forms in this chapter include the following:

1. Client information form (Figure 3-2)
2. Referral form (Figure 3-3)
3. Physical rehabilitation evaluation form (Figure 3-4)
4. Orthopedic evaluation form (Figure 3-5)
5. Neurologic evaluation form (Figure 3-6)
6. Daily flowsheet form (Figure 3-7)
7. Referral letter (Figure 3-8)

Text continued on p. 30

NEW CLIENT INFORMATION

Owner's name	
Address	
Home phone	
Work phone	
Veterinarian/surgeon	
Date of surgery/injury	

PATIENT INFORMATION:

Name	
Breed	
Age	
Sex	
Color	
Spayed/neutered	

PATIENT HISTORY:

Rabies vaccination	
Past medical history	
Previous surgery	
Allergies	
Special diet/medication	
Previous activity level	
History of present illness	
Treatment since injury/surgery	
Owner's goals	

Figure 3-2 Client information form.

CANINE REHABILITATION CLINIC

Referral Form

Client_______________________________ Patient_______________________________ Date_______________

Breed_______________________________ Sex____________ Age______________ Weight_______________

Referring veterinarian/clinic:____

Clinical condition:____ Onset/Sx date:_____________

Special instructions/precautions____

____

____

____

____

____

Frequency and duration: ___________________________Times per day for ____________________________days

☐ Board until __________________________ Drop off: ☐ MWF ☐ Other___________________________________

☐ T/Th

Plan: ☐ Evaluate and treat

☐ Hot pack
☐ Cryotherapy
☐ Ultrasound
☐ Electrical stimulation
☐ Therapeutic exercise
☐ Hydrotherapy
☐ Gait training
☐ Massage
☐ Joint mobilizations
☐ Weight-bearing/weight shifts
☐ Passive range of motion
☐ Neuromuscular reeducation
☐ OTHER:____

DVM Signature____

Figure 3-3 Referral form.

PHYSICAL THERAPY INITIAL EVALUATION

Patient's name:	
Date:	

PHYSICAL EXAMINATION:

Skin/incisions:		Color/temp:	
Heart rate:		Respirations:	

POSTURE/GAIT:

General observation:				
Preop/injury lameness:	Walk:		Trot:	
Postop/injury lameness:	Walk:		Trot:	
Standing limb position:		Sitting limb position:		

Circumference (cm):	70% femur	80% humerus	Joint line	Other
Affected:				
Unaffected:				
Other:				

RANGE OF MOTION:

Joint(s): Aff/Unaff	Flexion	Extension	AB/adduction	Varus/Valgus	Other
Hip:					
Stifle:					
Hock:					
Shoulder:					
Elbow:					
Carpus:					
Other:					

PALPATION:

Forelimb	
Hind limb	
Spine	
Other	

SPECIAL TESTS:

Neurologic:	
Orthopedic:	
Functional:	
Other:	

Figure 3-4 Physical rehabilitation evaluation form.

Continued

TREATMENT:

Modalities:	Manual:	Therex:
Interferential current	Massage	Gait training
Neuro-muscular electrical stimulation	Joint mobilization	Aquatic
	Passive range of motion	Functional
		Swiss ball
Other stim		Foam roll
Ultrasound	Other:	Owner education
Ice		Protocol review
Heat		Other:
Other		

ASSESSMENT/GOALS:

Decrease pain	
Decrease edema	
Increase weight-bearing	
Independent home exercise program	
Return to previous function	
Other	

PLAN:

Return visit	
Call for follow-up	
Call DVM	
Other	

DVM Signature ______________________________

Figure 3-4, cont'd

CANINE ORTHOPEDIC REHABILITATION
EVALUATION FORM

Patient________________________________ Client________________________________ Date________________

Breed________________________________ Age___________ Sex___________ Date of Sx/onset_______________

Referring vet/clinic:____ Diagnosis______________________

History:____
____
____
____
Medications:____
Client's goals____
____
Functional mobility____
____

OBJECTIVE
Involved limb: RF LF RR LR

Range of motion:

Forelimb	**R**	**L**
Shoulder		
1. Flexion	________	/ ________
2. Extension	________	/ ________
Elbow		
1. Flexion	________	/ ________
2. Extension	________	/ ________
Carpus		
1. Flexion	________	/ ________
2. Extension	________	/ ________

Rear limb	**R**	**L**
Hip		
1. Flexion	________	/ ________
2. Extension	________	/ ________
Stifle		
1. Flexion	________	/ ________
2. Extension	________	/ ________
Hock		
1. Flexion	________	/ ________
2. Extension	________	/ ________
Other	________	/ ________

Visual inspection/palpation:______________________________
____
____
____
____

Pain score:
0 = No pain on palpation of joint
1 = Mild pain; palpation completed
2 = Moderate pain; palpation completed with obvious discomfort noted
3 = Severe pain; palpation not completed
4 = Pain too severe; restraint/sedation needed to palpate

Limb circumference:

Muscle mass:

	Affected__________	Unaffected__________
Femur length	______cm	________cm
70% of length	______ cm	________cm
Girth	______ cm	________cm

Joint effusion:

	Affected__________	Unaffected__________
Patellar tendon	______ cm	________cm
2" above	______ cm	________cm
2" below	______ cm	________cm

Figure 3-5 Orthopedic evaluation form.

Continued

Gait analysis:

Orthopedic lameness

Degree of lameness (stance) ______	Degree of lameness (walk) ______	Degree of lameness (trot) ______
0 = Normal stance	0 = No lameness/weight-bearing on all strides observed	0 = No lameness/weight-bearing on all strides observed
1 = Slightly abnormal stance (partial weight-bearing)	1 = Mild subtle lameness with partial weight-bearing	1 = Mild subtle lameness with partial weight-bearing
2 = Moderately abnormal stance (toe-touch weight-bearing)	2 = Obvious lameness with partial weight-bearing	2 = Obvious lameness with partial weight-bearing
3 = Severely abnormal stance (holds limb off the floor)	3 = Obvious lameness with intermittent weight-bearing	3 = Obvious lameness with intermittent weight-bearing
4 = Unable to stand	4 = Full non-weight-bearing lame	4 = Full non-weight-bearing lame

Post-operative limb use

Degree of limb use (stance) ______	Degree of limb use (walk) ______	Degree of limb use (trot) ______
0 = Normal stance	0 = No lameness	0 = No lameness
1 = Slightly abnormal stance (partial weight-bearing)	1 = Lame but weight-bearing on >95% of strides	1 = Lame but weight-bearing on >95% of strides
2 = Moderately abnormal stance (toe-touch weight-bearing)	2 = Lame but weight-bearing on >50% and <95% of strides	2 = Lame but weight-bearing on >50% and <95% of strides
3 = Severely abnormal stance (holds limb off the floor)	3 = Lame but weight-bearing on >5% and <50% of strides	3 = Lame but weight-bearing on >5% and <50% of strides
4 = Unable to stand	4 = Non-weight-bearing lame or weight-bearing on <5% of strides	4 = Non-weight-bearing lame or weight-bearing on <5% of strides

Gait deviations:____
____
____
____

ASSESSMENT____
____
____
____

PLAN OF CARE____
____
____
____

Therapist Signature______________________________

Figure 3-5, cont'd

CANINE NEUROLOGIC/SPINAL REHABILITATION EVALUATION FORM

Client________________________________ Patient________________________________ Date________________

Breed________________________________ Age___________ Sex___________ Date of Sx/Onset_______________

Referring vet/clinic:____ Diagnosis______________________

History:____

____

____

Prior surgeries:____

Follow-up visit:________________________ Medications:____

Client's goals:____

____

OBJECTIVE

Involved area(s) : RF LF RR LR Spine__________

Mental status: Alert Depressed Stuporous Comatose Aggressive Fearful Disoriented

Palpation/postures:____

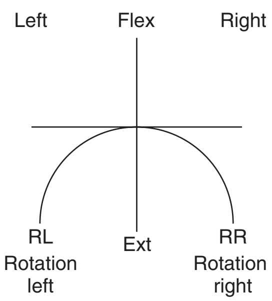
____

____

____

____

Active movements:



KEY:
Affected Body Segment
X Tender
O Center of pain
☰ Spasm
/// Guarding
* Reflex contraction
∞ ROM

Limb circumference:

Muscle mass:

Affected___________________		Unaffected__________
Femur length	______cm	__________cm
70% of length	______cm	__________cm
Girth	______cm	__________cm

Figure 3-6 Neurologic evaluation form.

Continued

Gait analysis:
Involved limb(s):______________

Degree of deficit (stance) ________________________

Degree of deficit (walk) ________________________

Degree of deficit (trot) ________________________

5 = Normal strength and coordination
4 = Can stand to support; **minimal paraparesis** and ataxia
3 = Can stand to support but frequently stumbles and falls; **mild paraparesis** and ataxia
2 = Unable to stand to support; when assisted, moves limbs readily but stumbles and falls frequently; **moderate paraparesis** and ataxia
1 = Unable to stand to support; slight movement; when supported **severe paraparesis**
0 = Absence of purposeful movement; **paraplegia** or **tetraplegia**

Deviations:____
____
____
____

Proprioception: (+ intact; – absent)

RF	LF	RR	LR

Spinal reflexes:
Key: 0 = Absent
+1 = Depressed
+2 = Normal
+3 = Exaggerated
+4 = Clonus
NE = Not examined

Crossed extensor: ______ Forelimb
+ Present ______ Hindlimb
– Absent

Forelimb	RF	LF	RR	LR
Flexor reflex				
Biceps (C6-C8)				
Triceps (C7-T2)				
Ext carpi rad (C7-T1)				
Deep pain				
Hindlimb				
Flexor reflex				
Patella (L4-L6)				
Cranial tibial (L6-L7)				
Gastrocnemius (L7-S1)				
Sciatic (L6-S1)				
Perineal (S1-S3)				
Deep pain				
Cutaneous trunci				

ASSESSMENT____
____
____

PLAN OF CARE____
____
____

Therapist Signature____________________________________

Figure 3-6, cont'd

CANINE ORTHOPEDIC REHABILITATION FLOWSHEET

PATIENT NAME:__________________________ CLINICAL CONDITION:________________ SX DATE:_________

CLIENT NAME:____

TIME	Date	Date	Date	Date	Date

Reassessment: Date / /

ROM Shoulder R________ L________ Hip R________ L________

Elbow R________ L________ Stifle R________ L________

Carpus R________ L________ Hock R________ L________

Limb circumference
Affected limb __________ Unaffected limb __________

Weight-bearing status:______________________________
Gait analysis:____________________________________

Proprioception:

RF LF RR LR

OTHER:____________________

PLAN:____________________

Reflexes: R L
Forelimb flexor
Rear limb flexor
Patellar

Therapist Signature__________________________________

Figure 3-7 Daily flowsheet form.

CANINE REHABILITATION CLINIC

{Date}

{Address}

Dear Dr. { },

Thank you for referring “Star” to Physical Therapy at The Canine Rehabilitation Clinic and Veterinary Hospital. Star has been treated for 3 days with ultrasound to help resolve the contracture of the digits of the right forelimb and the underwater treadmill to help regain full function of the left hindlimb. We are encouraging Star to use the leg in a buoyant environment to avoid exacerbating the contracture; second, the thermal effects of the water will help with improving joint mobility, and finally, the resistant property of the water will help to strengthen the muscles of the limb. I have also reiterated to her owner, Diane, to take Star for short walks (no more than 10 to 15 minutes at a time) no more than 2 to 3 times per day (on the off days from physical therapy).

Dr. Smith has left a message outlining the rehabilitation process and changes in medications to avoid further GI upset for Star. So far, we have seen no adverse reactions to the medications.

Again, thank you for this referral. We will continue physical therapy for 2 days this week and will provide the owner with a home care program to continue strengthening activities for Star. We expect that she will have progressive improvement.

Please do not hesitate to call if you have any questions or concerns: {Phone number}.

Sincerely,

Figure 3-8 Referral letter.

REFERENCES

1. *Healing the generations: a history of physical therapy and the American Physical Therapy Association*, Lyme, Conn, 1995, Greenwich Publishing Group.
2. The guide to physical therapist practice, ed 2, *Phys Ther* 81:21-22, 2001.
3. Physical Therapy Workforce Data. https://www.apta.org/WorkforceData/, 2013.
4. About the American Veterinary Medical Association. Who we are. https://www.avma.org/about/WhoWeAre/Pages/default.aspx, 2013.
5. North American Veterinary Technician Association: A regulated profession. http://www.navta.net, 2001.
6. Veterinary Technician Specialties. http://www.avma.org/animal_health/brochures/careers/technology_brochure.asp#specialties.
7. *A normative model of physical therapist professional education: version 2004*, Alexandria, VA, 2004, American Physical Therapy Association.

SECTION II

Basic Science of Veterinary Rehabilitation

4

Canine Behavior

Julie Albright

Optimal benefits of physical rehabilitation are achieved with a willing and relaxed patient. An animal that has enjoyable experiences and trusts the handler is more inclined to comply with rehabilitation procedures, which may result in greater success. For example, muscle contraction that occurs in a tense animal is contraindicated during passive range-of-motion exercises.[1] Similarly, a patient's avoidance of aversive handling could interfere with the detection of pain, an indicator of potential tissue damage during cryotherapy[2] or neuromuscular electrical stimulation sessions.[3] A physical rehabilitation session can subject an animal to potentially painful or frightening procedures that may result in aggressive behavior. Therefore making physical rehabilitation a pleasant experience for the animal can also decrease the risk of injury to the therapist and pet. Pet owners are more inclined to continue rehabilitation if their animals do not appear distressed or hesitant about the facility or personnel. Interpreting body postures and vocalizations in our patients is essential in recognizing fear and stress. Therefore this chapter begins with a basic review of canine and feline communication. Humane handling and behavior modification techniques to decrease fear and ensure a willing patient are also discussed.

Canine Communication

Body posture and facial expressions are usually the first clue to a dog's emotional state. Overt postures such as stiff, forward body position with erect ears and tail signal alertness and offensive aggression (Figure 4-1, dog 6). An offensively aggressive dog's jaw is squared and only front teeth are exposed (Figure 4-2, dog 3). Conversely, a dog that is trembling, in a crouching position, or on its side with its ears and tail tucked is easily recognized as a fearful dog (see Figure 4-1, dogs 4, 5, and 9; Figure 4-3). A fearful or defensively aggressive dog is also lower to the ground compared with assertive dogs (see Figure 4-1, dog 8), often approaches and retreats, and has looser facial expressions that allow all the teeth to be visible (see Figure 4-2, dogs 8 and 9). More subtle behaviors such as panting, lip licking, and yawning can also indicate stress in a dog. During or between procedures, one may notice the dog sitting or lying very still with eyes closed. This is another sign of stress or conflict that may be mistaken for a state of sleepiness or relaxation. While performing potentially painful or threatening procedures, handlers should not only watch for overt body postures, such as lip lifting (snarling), but also less perceptible signals such as an immobilization of the body or a pause in panting. Both of these behaviors may indicate the dog has just become more aroused and aggression may follow.

Two actions often misinterpreted are the tail wag and piloerection (or "raising hackles"). Although the tail wag can indicate a friendly dog and piloerection can indicate an aggressive dog, this is an oversimplification. A dog with a loose body posture and low, loose wagging tail may indeed be friendly, but a dog with a stiff body and a rapidly wagging tail, especially if isolated to the tail tip, is probably agitated.[4,5] Piloerection is triggered by a sympathetic response and signifies a reactive animal that may be fearful or imminently aggressive. Some contend that hair raised across the entire dorsum may be indicative of the latter and piloerection concentrated on just the dorsal hips or shoulders is a more fear-motivated response.[5]

Vocalizations are also helpful indicators of a dog's mood. In general, harsh, low-frequency sounds are an attempt to make the receiver back away, and high-pitched tonal vocalizations are more friendly or appeasing in nature.[6] Barking in dogs can convey a wide range of emotions such as alarm, attention-seeking, distress, threat, or warning. A dog distressed by isolation will give a high-pitched bark, whereas alarm or aggressive barks are lower in frequency and harsher sounding.[7,8] Aggressive barks are usually given in short, rapid sequence.[8] Growling occurs in aggressive and nonaggressive contexts among dogs, but when targeted to humans, growling is typically a low-level threat that can be indicative of defensive or offensive

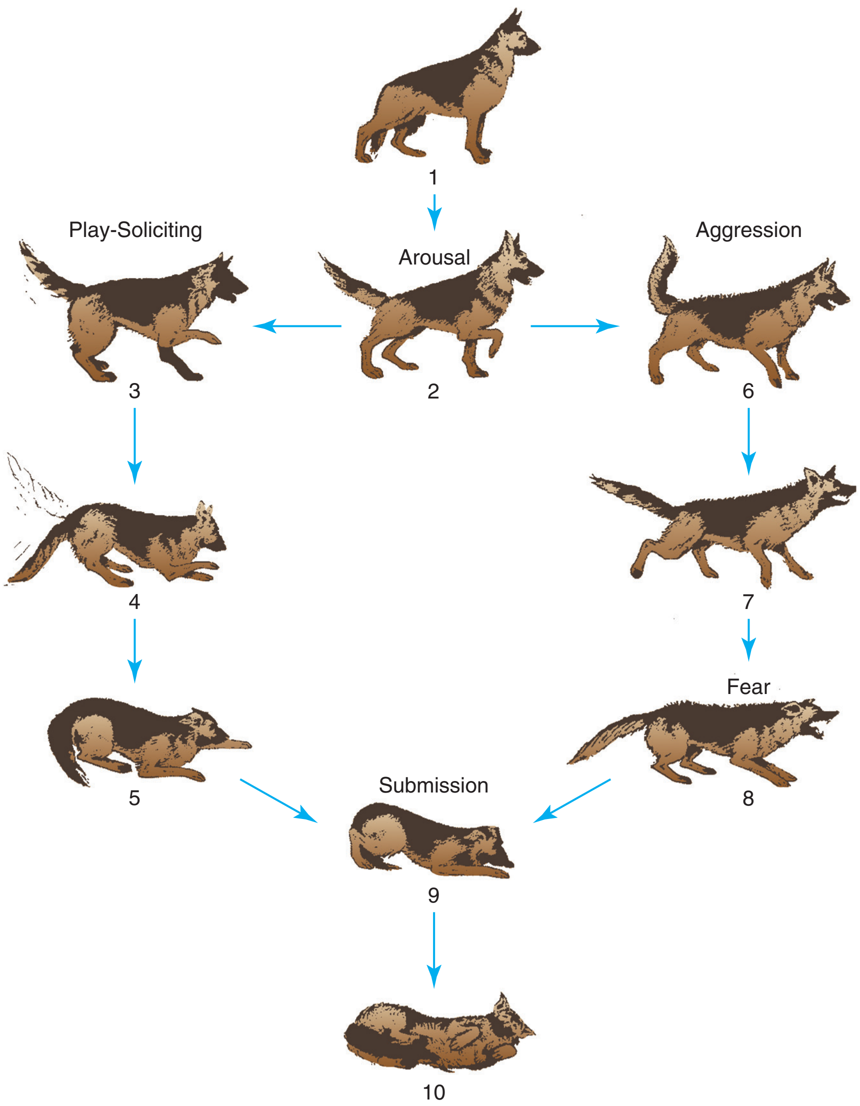


Figure 4-1 General body postures of dogs. Dog 1 shows a relaxed dog. Dog 2 is alert. Dog 3 shows playful behavior. Dogs 4 and 5 show increasing fear and submission. Dog 6 displays offensive aggression. Dog 7 shows mixed motivations of offensive and defensive aggression. Dog 8 shows defensive aggression. Dogs 9 and 10 show fear and/or submission. (From Overall KL: *Clinical behavioral medicine for small animals,* St Louis, 1997, Mosby.)

aggression.[9] A whine has cyclic frequencies and, in adult dogs, is typically made when the dog is in pain or frustrated.[10] Adult dog yelps are barklike and signify similar motivations as whines when directed at humans. Yelps are often heard during situations of greeting, submission, and play solicitation.[11] Howling is more common in nondomestic *canids* species such as wolves and coyotes and occurs as a long-distance call among pack members as a group vocalization to warn an approaching stranger, or possibly to promote pack affinity.[12] In domestic dogs, howling may be a response to a sound similar to a howl (e.g., siren or singing owner) and may represent a group vocalization.[11,12] In a physical rehabilitation facility, barks or howls may occur as a distress behavior when the dog is isolated in a cage or run. Barks are most commonly excited or aggressive displays in response to the presence of people or other animals. The motivation to vocalize should be reduced to decrease distress in the dogs and also the damage to human and patient hearing.

Dogs have an amazing sense of smell; thus urine and other olfactory stimuli are probably the most salient form of communication to the canine sensory system. In contrast, they may be the least recognized by humans in interspecies communication with dogs. In the rehabilitation facility, a dog that urinates, defecates, or expresses anal sacs, especially while displaying fearful body postures, is very distressed or excited. As a territorial behavior, some dogs may feel the need to urine mark, especially vertical surfaces previously urinated on by other dogs. In the clinic setting, stress may also be a contributing factor.

Feline Communication

As with dogs, a cat's vocalizations may help us to recognize its inner state.[10,13] The most common and variable vocalization is the meow and is given to seek attention or greet a social partner. A soft, rhythmical murmur is also given during greeting and social interactions. Purring is a

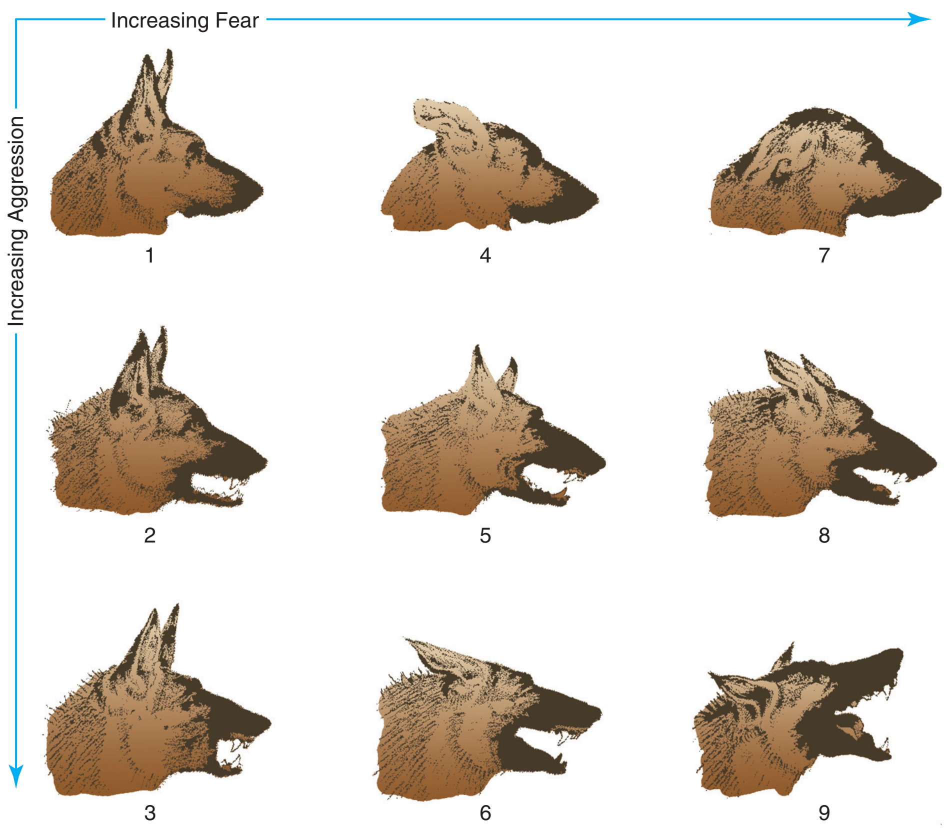


Figure 4-2 Facial postures of dogs. Figures from left to right show increasing fear. Figures from top to bottom show increasing aggressive motivation. Dog 1 is an alert dog. Dog 3 is offensively aggressive. Dog 7 is fearful and/or submissive. Dog 9 is defensively aggressive. All others are intermediate in fear and/or aggression. (From Overall KL: *Clinical behavioral medicine for small animals,* St Louis, 1997, Mosby.)

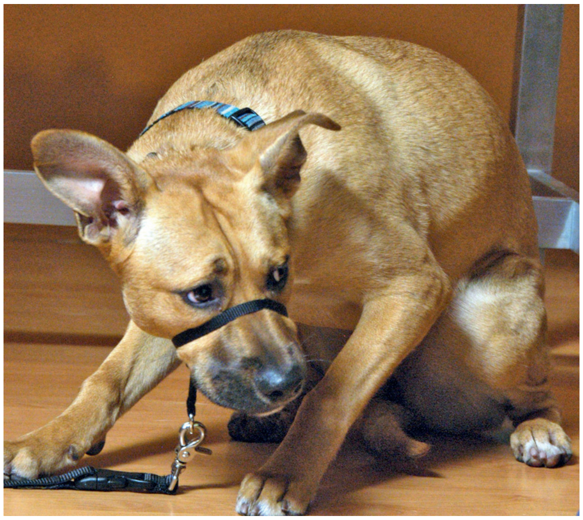


Figure 4-3 Fearful dog crouching, furrowing brows, and averting gaze. (Photo used by permission from Yin S: *Low stress handling, restraint and behavior modification of dogs and cats,* Davis, Calif, 2009, Cattle Dog Publishing.)

unique *felid* sound that is made by rapid contractions of the laryngeal muscles, innervated by a free-running neural oscillator that contracts and releases every 30-40 milliseconds.[14,15] People often assume a purring cat is a content cat, but this behavior can also occur in an anxious, painful, or sick cat. A cat purring during these stressful times is perhaps trying to solicit attention just as the purr from a very young kitten solicits attention from its queen. Aggressive cats may produce loud, harsh howls and growls, as well as hisses and spits. The growl is lower in frequency and longer in duration than the howl. The hiss is an open-mouthed sound invariably displayed in a defensively aggressive cat. During a painful event, a cat may give a loud, quick, high-pitched shriek, probably intended to startle the attacker into release.[16]

Emotions can also be visually communicated through body and facial postures.[5,10,16] A cat with an erect but curved tail, head up, and relaxed ear position is displaying a friendly greeting. Conversely, a cat with a vertically raised, piloerected tail with an arched back, and ears swiveled down or flattened against the head is defensively aggressive. As a cat becomes more frightened, its tail moves into a concave position or is tucked between the legs. An offensively aggressive cat's tail is lowered, but stiff and lashing. The body appears to be slanting down from tail to head and erect ears rotated back so the inside of the pinna is visible. An offensively aggressive cat's pupils are constricted but become more dilated with a feeling of fear and defense.

Patient Fear and Aggression during Rehabilitation

In a physical rehabilitation or veterinary setting, fear is the predominant motivation for aggression. Most animals show clear signs of fear and defense, but some may appear offensively aggressive. It is tempting to ascribe "dominance" to these animals, but on closer observation one usually sees the subtle and often fleeting signs of defensive aggression such as alternation of approach and retreat, gaze aversion, and ears shifting back. The dog motivated by fear often adopts more assertive postures when confronted with fear-producing stimuli, especially if the aggression successfully drove away the stimulus during previous encounters.

Dominance is a description of a relationship between individuals in which agonistic encounters (aggression and submission) establish who has priority to valued items, such as food or mates.[17,18] Using dominance as a static attribute of an individual's personality is incorrect. Dominance and territorial aggression are highly unlikely to be motivators for pets in a therapy facility. This is because they do not have an extended social relationship with personnel, nor have they likely spent enough time in the facility to become territorial. However, frustration and arousal caused by confinement can trigger cage or kennel aggression in an animal distressed by the sights, sounds, and smells of a facility. This is often misidentified as territorial aggression. Fear is the most reasonable impetus for aggression considering the circumstances.

Irrespective of an animal's emotional state, forceful or pain-inducing handling of animals is never warranted and is usually counterproductive.[18,19] Harsh methods of restraint can escalate a fear response to fear aggression, causing injury to the pet and handler. Once a pet has a fearful or aggressive response in the facility, subsequent visits will result in an even more extreme reaction, thus preventing the therapist from accurately assessing and performing treatments or exercises. Furthermore, owners will be less willing to bring the animal to the clinic for future treatments because of the animal's behavior and welfare. Clients will be more likely to endure the inconvenience of bringing a pet to rehabilitation if they feel the pet is being handled humanely and the pet seems to enjoy trips to the clinic.

Learning and Behavior Modification

Management of patient behavior begins with an understanding of how an animal learns. Learning is basically how consequences influence behavior and is broadly categorized as *nonassociative* or *associative*.[20] Nonassociative learning can be further categorized into *habituation* or *sensitization. Operant conditioning* and *classical conditioning* are the two types of associative learning.

Habituation occurs when an animal learns not to respond to an inconsequential stimulus on repeated exposures. *Flooding* is a term often used to describe exposing an animal to a high-intensity stimulus in the hope that the animal will stop reacting. Instead of habituation, *sensitization,* or an increase in the animal's reaction to a stimulus, may result. The nature of the stimulus (e.g., loud noise) and the level of stress the pet is experiencing are important factors in determining whether the result will be habituation or sensitization. Furthermore, an animal that is suffering from prolonged distress may experience *dishabituation* of stimuli and react fearfully toward objects or people to which it previously seemed indifferent.[21]

An example of nonassociative learning is a nervous dog that may be hesitant when urged to enter an underwater treadmill. The human handler detects nothing frightening about the apparatus so gently forces the dog into the treadmill with the expectation it will "just get over it" (i.e., flooding in an attempt to habituate). However, the stress the dog is experiencing may cause sensitization to the treadmill, handler, or even generalize to the entire facility, resulting in a dog that balks more vehemently and potentially becomes aggressive each time it is near the treadmill.

In this example, the dog has associated the surrounding stimuli such as the staff or room containing the treadmill with unpleasant physiologic effects of fear previously experienced in the treadmill. In this *classical conditioning* (or *Pavlovian learning*), a neutral stimulus is paired with a stimulus that naturally triggers a physiologic response (such as fear, pleasure, or excitement). After repeated pairings, the neutral stimulus alone involuntarily evokes the same response. Conversely, operant *conditioning,* also termed *instrumental learning,* is the voluntary performance of a behavior based on previous consequences. For example, an animal will try to repeat a behavior that results in a reward (*reinforcement*) and avoid those that cause unpleasant sensations such as pain or fear (*punishment*). Most animal training is based on this type of conditioning.

Reinforcement or punishment can be further qualified as negative or positive depending on whether something was applied or removed, not the emotion of the animal or trainer. In other words, a positively reinforced behavior is one repeated because the animal previously received something desirable immediately after performing that action. A behavior may also recur because the animal has learned that action will cause something aversive to be removed. This is negative reinforcement. A head halter is often used to decrease pulling on a leash (see Humane Control and

Safety Devices later in this chapter for more detailed description). Walking close to the handler is negatively reinforced because the pressure on the nose loop is released when the dog stops pulling and he learns that remaining near the human handler prevents the nose pressure. This is often confused with positive punishment, or decreasing the likelihood of a behavior by inflicting something aversive on the animal after it has performed an undesirable behavior. In the same example, the pulling behavior decreases as the walk proceeds because the dog learns jerking ahead of the handler exacts pressure on the nose. Pulling is positively punished, and remaining near the human is negatively reinforced. A behavior can also become less likely through negative punishment, or removal of something the animal considers rewarding when it performs an unwanted behavior. Most dogs consider their owners' affections enjoyable so removing this stimulus when the dog is barking or jumping for attention decreases this pestering behavior. The best result comes from well-timed positive reinforcement (giving attention) when the dog is behaving appropriately (sitting) in addition to the immediate negative punishment (removal of attention) of the unruly behavior (jumping).

Success of behavior modification depends on the timing and relevancy of the behavior with the punisher or reinforcer. The stimulus must be applied within just a second or two of the animal's action and be applied consistently for the animal to be able to predict which behaviors elicited the pleasant or unpleasant consequences. Inappropriately applied rewards can result in problems such as rewarding the incorrect behaviors and potentially increasing anxiety. However, poorly applied punishment does not just result in poor efficacy, but also becomes a welfare concern. The inability of an animal to predict and control when it will receive something that induces fear or pain quickly creates a distressed animal. Animals that perceive an aversive event as unavoidable may become apathetic and not attempt escape, a phenomenon called *learned helplessness*.[22] This indifference could be mistaken for relaxation, but short- and long-term activation of the stress neuroendocrine system may lead to compromised health and mental well-being in therapy patients.

Positive punishment is often the technique first employed to try to change an animal's unwanted behavior. However, for the reasons previously discussed, this is rarely appropriate in a clinic or therapy setting. Subjecting an animal to discomfort or verbal reprimand that induces an unpleasant emotional state may suppress behavior (operant conditioning), but also classically conditions fear of the facility and staff. The use of positive reinforcement, negative punishment, and environmental management (e.g., quiet facility and training of staff to correctly read subtle animal body postures) is strongly recommended in a clinic to prevent fear and escape behaviors. The inconvenience, dietary restrictions of some patients, and bias against "bribing" our pets for certain behaviors often limit the use of food in the clinic. Anything the pet finds pleasurable (toy, presence of owner, an activity like "fetch") can be used as a reward, and thus given as a reinforcer or removed as a punisher. In reality, however, food is an intrinsic need and the most effective, consistent motivator for our patients. Clients can be instructed to consult with a reputable behaviorist or trainer[23] regarding the importance of setting behavior criteria and reward schedules to avoid "bribing" for good behavior. But this is of little concern in the clinic because maintaining a relaxed emotional state in an environment fraught with novel and potentially frightening experiences requires a large "pay check" for the pet. If obesity is a concern, the owner can adjust the diet at home to ensure the appropriate caloric intake. Unless a patient cannot be fasted, the owner should be instructed to feed the animal a maximum of half the morning meal; very small treats are provided during the sessions so the pet remains motivated to take the food. A variety of very small treats should be kept on hand in each therapy room. Common favorite foods are real meat or fish, semimoist treats, or sticky foods like peanut butter, meat-flavored baby food, or flavored pastes in nozzle cans.*

The food can be used to *countercondition* or change the animal's emotional response to the target that elicits fear. As a form of associative learning, counterconditioning can be classical or operant in nature. Classical counterconditioning involves simply providing the dog with rewards in the presence of the fear-producing target with the expectations of replacing the fear response with a pleasant emotion. Often the outcome is more successful if the dog is asked to perform a previously trained command that is incompatible with the fear response while in the presence of the stimulus (operant counterconditioning). Again using the underwater treadmill as an example, a dog that is trembling or attempting to flee as the water fills the chamber could be rewarded for moving forward as the water flow begins by targeting this skill that has been previously mastered outside the apparatus (see Handling of the Physical Rehabilitation Patient later in this chapter). In both classical and operant counterconditioning protocols, the emotional state is being changed. Frequently the target produces such a level of stress that getting the animal focused on a command or a reward may be very difficult. Thus most behaviorists advocate a *systematic desensitization* protocol in which the animal is habituated to a very low stimulus level (e.g., expose the dog to the treadmill at great distance), and then the intensity level is gradually increased (the dog is brought closer to treadmill) with the goal of never triggering fear. Typically a dog is asked to perform commands for treats as the distance is decreased. This is called a *desensitization/counterconditioning protocol* and is the crux of most behavior modification.

*Kong Easy Treat, Kong Company, Golden, Colorado.

Setting the Stage

Preventing anxiety rather than trying to calm a highly distressed dog is a more successful strategy in maintaining a compliant patient. This begins long before the dog enters the treatment room. A dog that has car distress or motion sickness on the way to the clinic is more likely to be stressed during the session. The dog's regular veterinarian should be consulted regarding antiemetic medications if nausea or vomiting during the car ride is a consistent problem. Other strategies to decrease car distress include desensitization to the car itself and then the motion of riding in the car.[18] Visual restriction (e.g., Calming Cap* as shown in Figure 4-4), stabilization via a crate or seatbelt, as well as pheromone therapy[24,25] and lavender aromatherapy[26] can have a calming effect on dogs showing excitable or stress behaviors in the car.

The reception area and therapy rooms should be designed to minimize stimuli that can cause excitement. The waiting room can be made less stressful by instructing staff to move slowly and minimize noise. Arranging seating and providing solid structures in the sitting area so fearful or reactive animals can be visually isolated can be very helpful. Visual sequestration is also recommended for the cage and kennel area as well. Cats should always be provided with hiding places and dogs in cages and larger runs may also feel more secure if a partial visual barrier is provided.

First impressions with staff are important in gaining the patient's trust. Quickly approaching, leaning over, and reaching out are considered threating actions to a dog (Figure 4-5, *A*). Instead, the staff member should provide enough time and space for the dog to acclimate to its new surroundings (Figure 4-5, *B*). A crouching or sitting human who is avoiding eye contact is much less threatening. Treats can be gently rolled to the dog from a distance as

*Calming Cap, Premier Pet Products, Midlothian, Virginia.

Figure 4-4 A Calming Cap can decrease visual stimulation and stress during car rides. (Photo courtesy of Thunderworks, Durham, North Carolina.)

Figure 4-5 **A,** Incorrect greeting: the staff member leans and reaches over the dog. The dog perceives this action as a threat and reacts with fearful body postures and avoidance of the approaching person. (Photo used by permission from Yin S: *Low stress handling, restraint and behavior modification of dogs and cats,* Davis, Calif, 2009, Cattle Dog Publishing.) **B,** Correct greeting: approach dog backwards or from side in a kneeling position and offer treat by slowly extending forearm out to the dog. Allow the dog to approach instead of cornering it. (Photo used by permission from Yin S: *Low stress handling, restraint and behavior modification of dogs and cats,* Davis, Calif, 2009, Cattle Dog Publishing.)

the staff member greets and gets a history from the owner. The dog is less fearful if encouraged with treats and praise to voluntarily approach the unfamiliar person. The dog may feel trapped and defensive if the unfamiliar person invades its personal space. After the animal is in the therapy area, staff members can continue to make the visit enjoyable by speaking to the dog in an upbeat voice and luring it around the facility with small food treats.

Animal facilities have an abundance of smooth, cleanable surfaces that can amplify sound waves. Barking in dogs can easily reach the 90-decibel limit of the Occupational Safety and Health Act.[27] Human ears can be protected with safety devices, but the welfare of our patients is still compromised if noise is not reduced. An architect familiar with noise-reducing design and building materials should be consulted for new facility construction.[28] Although retrofitting existing facilities to reduce noise and visual contact can be more challenging, cleanable but sound-absorbing rubber and certain fiberglass panels can be used in most areas.

Classical music may promote quiet and resting behavior, whereas heavy metal music seems to agitate dogs.[29] Barking and other anxiety-related behaviors may also be decreased by occupying the dog with treat-dispensing objects (e.g., Kong toys* or Premier Pet Products Busy Buddy line of toys†) filled with palatable food. Decreasing the motivation to bark by alleviating underlying distress is preferable to aversive methods that may increase anxiety, but persistent barking by just one or two dogs can increase the stress level of all other patients and staff. A spray (citronella or scentless) bark collar for specific offenders may be helpful in these situations.[30]

Pheromones are another simple modality that may be used to reduce stress in kennel and therapy areas.[31] Feliway is a synthetic version of the feline facial pheromone. The diffuser can be used in wards or treatment rooms, and the spray form can be applied to towels or hands several minutes before the procedure. Adaptil‡ is a synthetic version of a lactating bitch mammary pheromone and is also available as a diffuser or spray. The spray reapplied every few hours to a towel or bandana could potentially provide better pheromone concentration and efficacy for that individual dog or cat. Collars that are impregnated with pheromones and claim to emit the chemical for approximately 1 month are also available (Adaptil‡ and Nurturecalm§).

Finally, nonslip flooring and substrates are essential to creating a relaxed patient. A dog that feels unsteady is more likely to panic and injure itself or staff during restraint, and become an uncooperative participant. Thick rubber mats afford both stable footing and cushioning of pressure points that may become uncomfortable during recumbent procedures. Small dogs and cats should be placed on a soft but secure towel on the examination table.

Humane Control and Safety Devices

Appropriate control over the dog helps to prevent unruly behavior, such as pulling, jumping, or aggressive lunging. Head halters are an excellent tool and are underused in the clinic setting. There are several styles and brands of head halters with slightly different designs, but all have two loops—one behind the ears and one around the nose. With most designs the leash attaches under the jaw. This provides better control because the dog's pulling away from the handler puts pressure on the nose strap and pulls the nose back to the handler. Although the pressure is aversive, it is safe and not painful when applied correctly. The owner should be encouraged to acclimate the dog to the head halter at home by associating it with treats and other enjoyable activities, like walks (Figure 4-6). However, this can usually be done in just a few minutes at the clinic. Head halters can be extremely useful during the session. A dog that becomes distressed will often attempt to escape from the situation. The handler may, in turn, react by speaking sharply to the dog or taking a "death grip" on the patient. This reaction may suppress the struggle in that moment but it will only increase the patient's fear and make subsequent interactions more difficult. Some dogs may appear calmer when simply wearing a properly fitted head halter. If a dog does become overly aroused, judicious use of pressure and release of pressure from the nose loop can quickly calm the dog by preventing escape and allowing the handler to remain calm and quiet.

A head halter can prevent low levels of aggression, but dogs with a biting history may require a muzzle. A basket muzzle is preferred in most situations. A basket muzzle allows the dog to take treats and pant, whereas a sleeve muzzle can exacerbate distress by restricting panting and breathing. Although there may be no obvious difference for brief examinations or events like a venipuncture, extended procedures are more common in physical rehabilitation, thus making basket muzzles more desirable. As with the head halters, animals will accept the muzzle more willingly if a desensitization program is outlined for the owners and performed before it is used in the session. In short, place spreadable food such as peanut butter, canned cheese, or meat-flavored dog pastes (e.g., Kong liver flavor paste) into the muzzle. Encourage, but do not force the dog to put its nose in the muzzle for the treats (see Figure 4-6, *A*). When the dog does not hesitate to put its nose in the muzzle and keep it there, the strap behind the ears can be fastened. The time the dog wears the muzzle can be gradually increased by a few seconds as long as it

*Kong Rubber Toys, Kong Company, Golden, Colorado.
†Busy Buddy Toys, Premier Pet Products, Midlothian, Virginia.
‡Adaptil, Ceva Sante Animale, Rutherford, New Jersey.
§Nurturecalm 24/7 Pheromone Collar, Meridian Animal Health, Omaha, Nebraska.

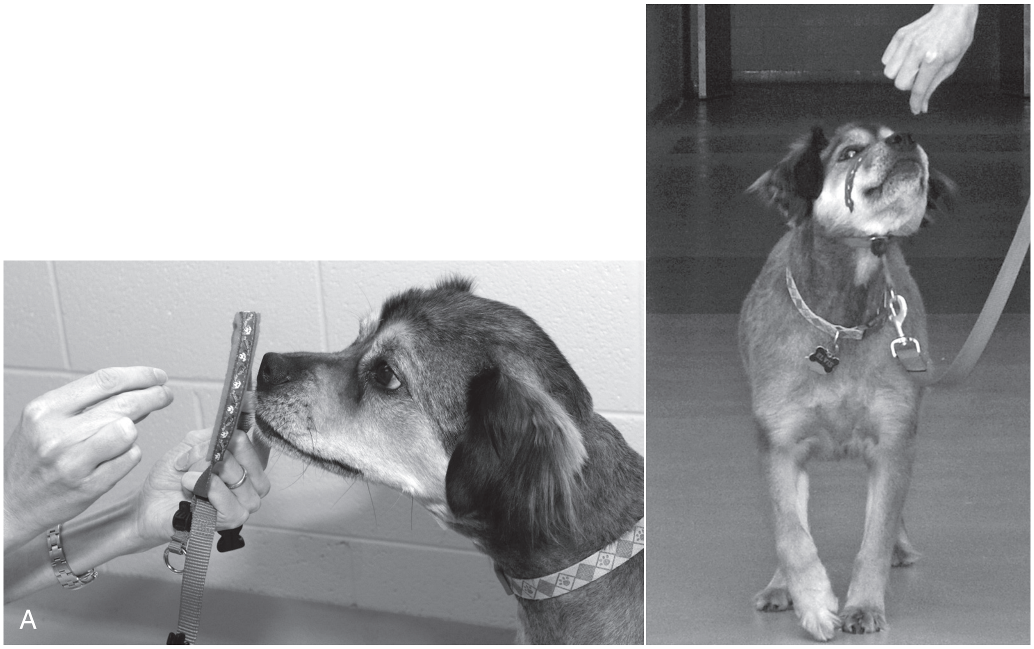


Figure 4-6 **A,** Acclimate the dog to the head halter by initially rewarding with treats when he puts his nose through the loop. **B,** Continue to associate the head halter with rewards by treating the dog for calmly walking by your side.

is relaxed. Continue to provide the treats while the dog is wearing the muzzle.

Handling of the Physical Rehabilitation Patient

Physical therapists and their staff are encouraged to read, watch videos, and practice handling techniques that are extensively detailed in other publications to supplement the material in this chapter. In *Low stress handling, restraint, and behavior modification of dogs and cats,*[18] Yin outlines 10 basic principles of low-stress handling, including proper body support, decreasing fear to reduce patient struggling, and minimizing the duration of restraint. Most importantly, handlers should become highly skilled at recognizing subtle signs of fear and distress in pets. When fear is first detected, handling strategies can then be adjusted, allowing the animal to remain relaxed and avoid extreme escape behaviors and aggression.

Choke chains and slip leads should be avoided. As discussed previously, head halters provide the most humane control. Treats and the head halter can be used to lure the dog from one position to another and minimize the need for physical positioning. However, the use of hands and body to properly place the animal is usually necessary at some point during a therapy session. When moving a dog from one position to the next, motions should be slow and smooth, and the handler's body should be a solid support, not painfully forcing the patient into a certain position. Staff should become adept at towel-wrap restraints for cats

Figure 4-7 Secure and proper towel-wrap restraint can allow access to both the head and hind-end while decreasing the cat's stress and injury to the handler.

so uncomfortable scruffing and stretching can be avoided (Figure 4-7). The tight wraps provide security and ample access to the needed area while also decreasing agitating sensory overstimulation.[18]

Figures 4-8 and 4-9 show how the head halter can be used to keep the dog focused on the reward during a procedure. Note the point of control is on the dog's head. It would be difficult for him to turn his mouth far enough to bite; however, the body restraint is minimal so any discomfort from the ultrasound may be easily detected. Figure 4-8 is appropriate for a dog with a history of aggression. The leash is kept between the dog and the wall, preventing the mouth from reaching the handler and therapist. An Elizabethan collar or rolled towel wrapped around the neck can also be used to block bites, particularly in cats and

Figure 4-8 The handler uses the head halter and treats to distract and control the dog during a procedure. This low-restraint method keeps the dog calm and also prevents the dog from turning to bite the handler or therapist. However, a basket muzzle can be placed over the head halter for added safety.

Figure 4-9 Restraint for examining the hind limbs or back of an aggressive dog. The dog is placed against a wall and one handler stands close to the dog's shoulder. This handler keeps the leash of the head halter between the dog and the wall, thus preventing the dog from turning to bite. Another handler is offering treats during the examination or procedure to encourage the dog to remain stationary. A basket muzzle may be placed over the head halter.

brachycephalic (very short muzzled) dogs for which a head collar can be difficult to fit. Also note that treats are given liberally throughout the procedure. Handlers often wait until after the procedure to reward the dog for good behavior. As discussed in Learning and Behavior Modification earlier in this chapter, the action the animal is performing within 1 second of receiving the reward or punishment is the behavior being operantly conditioned. Thus a treat and verbal praise given as the procedure is ending and restraint is removed is rewarding the pet for moving away from the handler. Although the procedure room and staff are possibly being associated with a pleasant emotion through classical conditioning, remaining calm and still *during* the procedure is not being reinforced.

Figure 4-10 Luring a dog into the water treadmill using a previously conditioned hand-targeting command. The dog mastered targeting in quiet areas and then near the treadmill before he was lured into the apparatus.

Tasty treats or other motivators like a favorite toy or person can be used to lure a hesitant dog onto apparatuses like inclines or treadmills. Asking the animal to perform a command for the reward often improves compliance by focusing the dog on an alternate activity. A correctly performed counterconditioning with systematic desensitization may take more time to complete during the initial training phase, but it is a time saver, as well as being more humane, over the long term. Figure 4-10 demonstrates a dog being lured into the water treadmill using a hand target command. The dog was previously taught to touch the hand with his nose when giving the visual cue of the hand and the verbal command "touch." This dog is nervous about approaching the treadmill. The targeting and other commands are initially asked at a great distance from the apparatus. The handler and dog close the distance to the treadmill by a foot or two when the dog appears relaxed and focused on the handler. If the apparatus has a potentially frightening sound, the first phase of the desensitization should be the gradual approach with the machine off and quiet. The next stage is repeating the session with the motor turned on. The dog should not be forced into approaching. A dog too frightened, tired, or unmotivated to ambulate on the treadmill is not receiving the full benefit of the exercise and may injure itself. Verbally praising the dog and luring with the target and treats increase the dog's motivation and classically condition the patient to the treadmill with an enjoyable emotion. This procedure can be used for a myriad of active exercises such as incline walking, swimming, dancing, and cart exercises. Target training and desensitization, along with good control using a head halter, can also successfully condition the pet to participate willingly in other exercises, such as the balance platform and exercise ball, both of which can be disconcerting because of the imbalance they create.